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United States Patent	12416642
Kind Code	B2
Date of Patent	September 16, 2025
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Composition and methods for measuring antibody dynamics

Abstract

Using protein structural probes one can identify tumor-induced or -produced (TIPS) factors that bind to therapeutic antibodies and change their dynamic structure, thereby negatively affecting their humoral immune functions as well as their pharmacologic activity. Using such protein structural probes and TIPS factors one can screen and identify inhibitors that can counter the binding of TIPS factors to affected therapeutic antibodies. These inhibitors can be used in the presence of a TIPS factor-susceptible antibody (TSA) for treating cancer. An inhibitor can be used alone or in combination with chemotherapy for treating cancer. Patients can be screened to identify those with low or no TIPS factor production as candidates for antibody therapy even in the case in which the antibody is a TSA. Conversely, those with high TIPS factor production are candidates for inhibitor therapy.

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Appl. No.:	17/277098
Filed (or PCT Filed):	September 18, 2019
PCT No.:	PCT/US2019/051706
PCT Pub. No.:	WO2020/068511
PCT Pub. Date:	April 02, 2020

Prior Publication Data

Document Identifier	Publication Date
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Related U.S. Application Data

us-provisional-application US 62737414 20180927

Publication Classification

Int. Cl.: **G01N33/68** (20060101); **C07K16/00** (20060101); **C07K16/28** (20060101); **C07K16/32** (20060101); **G01N33/542** (20060101)

U.S. Cl.:

CPC **G01N33/6854** (20130101); **C07K16/00** (20130101); **C07K16/2887** (20130101); **C07K16/32** (20130101); **G01N33/542** (20130101); G01N2500/02 (20130101); G01N2500/20 (20130101)

Field of Classification Search

CPC: G01N (33/6854); G01N (33/542); G01N (2500/02); G01N (2500/20); G01N (33/6857); G01N (2021/6441); G01N (33/574); G01N (33/57484); C07K (16/00); C07K (16/2887); C07K (16/32); C07K (2318/00); C07K (16/4208); C07K (16/4283)

USPC: 435/7.1; 435/7.9; 435/7.8; 436/501; 436/518

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a U.S. National Stage application under 35 U.S.C. § 371 of International Application PCT/US2019/051706, filed Sep. 18, 2019, which claims the benefit of priority to U.S. Provisional Application No. 62/737,414, filed Sep. 27, 2018. This application claims the benefit of the filing date of each of these prior applications. The content of each prior filing is expressly incorporated herein.

TECHNICAL FIELD OF THE INVENTION

(1) This invention is related to the area of humoral immuno-oncology. In particular, it relates to methods, kits, and compositions to improve antibody efficacy in inhibiting cancer growth or treating other diseases.

BACKGROUND OF THE INVENTION

(2) The human immune system is comprised of two major types of adaptive immunity, namely cellular and humoral immunity. With the recent discovery of immune checkpoint inhibitors mediating cellular immunity (Hodi F S, et al. N Engl J Med 363:711-723, 2010), the pursuit of agents that can block these pathways have gained significant knowledge on approaches to overcome this tumor survival mechanism. While several commercially approved therapeutic anti-cancer antibodies have been reported to exhibit their anti-tumor effects via the humoral-mediated antibody dependent cellular cytotoxic (ADCC) and complement dependent cytotoxic (CDC) immune pathways (DiLillo D J, Ravetech J V, Cancer Immunol Res 3:704-713, 2015), the understanding and discovery of potential humoral immune pathways that can be suppressed via tumor-induced or -produced factors (TIPS factors) has been less interrogated due to the unobvious mechanism(s) by which they can mediate immune suppression. The antibody-mediated humoral immune response is governed by the coordination of antibody-cell surface antigen engagement that in turn positions the antibody on the antigen epitope at a certain proximity to the cell surface. In cases where this positioning is optimal, cell surface bound antibodies may engage with Fc-gamma receptors on Natural Killer (NK) or myeloid/monocytic cells to initiate ADCC as well as engage with the C1q complement initiating protein to cause death of antibody-bound cells via the classical complement CDC pathway (Reuschenbach M, et al. Cancer Immunol Immunother 58:1535-1544, 2009). These effects have been observed during the development of several therapeutic antibodies such as rituximab, trastuzumab, cetuximab as well as a number of experimental antibodies (Zhou X, et al. Oncologist 13:954-966, 2008; Hsu Y F, et al. Mol Cancer 9:8, 2010; Spiridon C I, et al. Clin Cancer Res 8:1720-1730, 2002; Kline J B, et al. Eur J Immunol 48:1872-1882, 2018). Similarly, humoral responses have been shown to occur within a patient's own immune response to

dysregulated and/or aberrantly growing cells in response to vaccines and those with indolent disease, yielding antibodies predominantly of the IgM class with anti-proliferative as well as humoral immune-mediated killing activities (Staff C, et al. *J Clin Immunol* 32:855-865; Branden S, et al. *Cancer Res* 63:7995-8005, 2003).

(3) TIPS factors can be readily detected in patient sera using established assays, and therefore their levels can be conveniently monitored for the purpose of establishing prognosis and/or disease stage and recurrence. Recent findings have shown that TIPS factors such as MUC16/CA125 (referred to as CA125 herein) may suppress humoral immune responses by directly binding to antibodies and perturbing the Fc region, thus making it less effective at engaging with Fc-gamma activating receptors FCGR2A or FCGR3A on immune cells and/or complement-mediating proteins, including C1q. Several of these data come from clinical studies of anti-cancer antibodies that rely on immune-effector mechanisms for their pharmacologic activity. Serum CA125 levels have been found to correlate with clinical outcome in a number of reported cases. These include reports on the experimental farletuzumab antibody in ovarian cancer and amatuximab antibody in mesothelioma (Vergote I, et al. *J Clin Oncol* 34:2271-2278, 2016; Nicolaides N C, et al. *Cancer Biol Ther* 13:1-22, 2018). Several reports have also shown the correlation of serum CA125 levels and shorter progression free survival (PFS) in patients with Hodgkin's and Non-Hodgkin's lymphoma, whereby patients with follicular lymphoma treated with rituximab plus CHOP (cyclophosphamide, doxorubicin (hydroxydaunomycin), vincristine (Oncovin® prednisolone) had a 31.4% improvement in 5-year PFS when CA125 levels were in the normal range (Prochazka V, et al. *Int J Hematol* 96:58-64, 2012). Moreover, CA125 was reported to be a prognostic factor in patients treated with standard chemotherapy who showed a 26.3% improvement in 3-year survival when their serum CA125 levels were within the normal range as compared to those with CA125 levels above the normal range (Bailey O, et al. *Leukemia Lymph* 44:1733-1738, 2003). There is a continuing need in the art to develop tools for discovering and analyzing the role of TIPS factors, their effects on TIPS-susceptible antibodies and a means for countering their negative effects in humoral immunity in cancer patients.

SUMMARY OF THE INVENTION

(4) One aspect of the invention is a kit for characterizing an antibody comprising a Fab and a Fc domain. The kit comprises a first fluor-labeled protein or aptamer that specifically binds to the Fab domain of the antibody and a second fluor-labeled protein or aptamer that specifically binds to the Fc domain of the antibody. The first fluor and the second fluor participate in fluorescence resonance energy transfer (FRET) when bound to the antibody. Optionally, the kit further comprises the antibody.

(5) In one aspect of the invention a kit is provided for characterizing a test antibody comprising a Fab and a Fc domain. The kit comprises a first fluor-labeled protein or aptamer that specifically binds to the Fab domain of the test antibody, and a second fluor-labeled protein or aptamer that specifically binds to the Fc domain of the test antibody. The first fluor and the second fluor participate in fluorescence resonance energy transfer (FRET) when bound to the test antibody.

(6) In another aspect of the invention an antibody is provided that comprises a Fab and a Fc domain. The antibody is labeled with a first and a second fluor that participate in FRET. The first fluor is attached to an amino acid residue in the Fab domain and the second fluor is attached to an amino acid residue in the Fc domain.

(7) In one aspect of the invention a kit for characterizing an antibody is provided. The antibody comprises a Fab and a Fc domain. The kit comprises an antibody labeled with a first fluor; and a protein or aptamer labeled with a second fluor. The first and second fluors participate in FRET when the protein or aptamer binds to the antibody. The first fluor is attached to an amino acid residue in the Fab domain and the protein or aptamer binds to the Fc domain, or the first fluor is attached to an amino acid residue in the Fc domain and the protein or aptamer binds to the Fab domain.

(8) In yet another aspect of the invention a composition is provided that comprises an antibody and a first and second protein or aptamer. The antibody comprises a Fab and a Fc domain. The antibody is labeled with a first and a second fluor that participate in FRET. The first fluor is attached to an amino acid residue in the Fab domain and the second fluor is attached to a Fc-binding protein or aptamer.

(9) Another aspect of the invention, a composition is provided that comprises an antibody that comprises a Fab and a Fc domain. The antibody is labeled with a first and a second fluor that participate in FRET. The first fluor is attached to a Fab binding protein or aptamer and the second fluor is attached to an amino acid in the Fc domain.

(10) In still another aspect of the invention a method is provided for characterizing a dual-labeled antibody. A first fluor-labeled protein or aptamer and a second fluor-labeled protein or aptamer is contacted with the antibody to be characterized to form a ternary complex. The first fluor-labeled protein or aptamer binds to the Fab domain and the second fluor-labeled protein or aptamer binds to the Fc domain. The first and second fluors participate in FRET. Fluorescence resonance energy transfer (FRET) of the ternary complex is determined. The ternary complex is contacted with a tumor-induced or -produced factor (TIPS factor). FRET of the ternary complex in the presence of the TIPS factor is determined. The TIPS factor may be known or unknown.

(11) In a further aspect of the invention a method is provided for characterizing an antibody. FRET of a dual-labeled antibody which comprises a Fab and a Fc domain is determined. The dual-labeled antibody is labeled with a first and a second fluor that participate in fluorescence resonance energy transfer (FRET). The first fluor is attached to an amino acid residue in the Fab domain and the second fluor is attached to an amino acid residue in the Fc domain. The dual-labeled antibody is contacted with a tumor-induced or -produced factor (TIPS factor). FRET of the dual-labeled antibody in the presence of the TIPS factor is determined. The TIPS factor may be known or unknown.

(12) Another aspect of the invention is a method for characterizing a dual-labeled antibody. FRET of a dual-labeled antibody which comprises a Fab and a Fc domain is determined. The dual-labeled antibody is labeled with a first and a second fluor that participate in fluorescence resonance energy transfer (FRET). The first fluor is attached to an amino acid residue in the Fab or Fc domain and the second fluor is attached to a protein or aptamer that binds to the other domain. The dual-labeled antibody is contacted with a tumor-induced or -produced factor (TIPS factor). FRET of the dual-labeled antibody in the presence of the TIPS factor is determined. The TIPS factor may be known or unknown.

(13) In one aspect of the invention, a method is provided for screening test substances for the ability to mitigate the effect of a tumor-induced or -produced factor (TIPS factor) on a TIPS-Susceptible Antibody (TSA). A TIPS-susceptible antibody is contacted with (a) a first fluor-labeled protein or aptamer that specifically binds to the Fab domain of the TIPS-susceptible antibody, (b) a second fluor-labeled protein or aptamer that specifically binds to the Fc domain of the TIPS-susceptible antibody, and (c) a TIPS factor to form a first complex. Fluorescence resonance energy transfer (FRET) of the first complex is measured. A TIPS-susceptible antibody is contacted with (a) a first fluor-labeled protein or aptamer that specifically binds to the Fab domain of the TIPS-susceptible antibody, (b) a second fluor-labeled protein or aptamer that specifically binds to the Fc domain of the TIPS-susceptible antibody, (c) a TIPS factor, and (d) a test substance, to form a second complex. Fluorescence resonance energy transfer (FRET) of the second complex is measured.

(14) In another aspect of the invention a method is provided for screening test substances for the ability to mitigate an effect of a tumor-induced or -produced factors (TIPS factors) on a TIPS-susceptible antibody comprising a Fab and a Fc domain. The TIPS-susceptible antibody is contacted with the TIPS factor. The TIPS-susceptible antibody is labeled with a first and a second fluor that participate in FRET. The first fluor is attached to an amino acid residue in the Fab domain

and the second fluor is attached to an amino acid residue in the Fc domain. A first complex is formed. FRET of the first complex is measured. The TIPS-susceptible antibody is contacted with the TIPS factor and a test substance, forming a second complex. FRET of the second complex is measured.

(15) In another aspect of the invention a method is provided for screening test substances for the ability to mitigate the effect of a tumor-induced or -produced factors (TIPS factors) on a TIPS-susceptible antibody. A TIPS-susceptible antibody is labeled with a first and a second fluor that participate in FRET. The first fluor is attached to an amino acid residue in the Fab or Fc domain and the second fluor is attached to a binding protein or aptamer to the other domain. The dual-labeled test antibody is contacted with a TIPS factor to form a complex. The TIPS-susceptible antibody comprises a Fab and a Fc domain. The antibody FRET of the first complex is measured. A TIPS-susceptible antibody is contacted with a TIPS factor and a test substance, to form a second complex. FRET of the second complex is measured.

(16) In still one more aspect of the invention, a method is provided for characterizing an antibody or for characterizing a pair of proteins or aptamers that bind to an antibody. The antibody is contacted with a first protein or aptamer that specifically binds to the antibody in its Fab domain. The first protein or aptamer is attached to a solid support, to link the antibody to the solid support. The antibody linked to the solid support is contacted with a second protein or aptamer that specifically binds to the test antibody in its Fc domain. The second protein or aptamer is labeled with a detectable label. The amount of the detectable label linked to the solid support is determined. The antibody linked to the solid support is contacted with the second protein or aptamer that specifically binds to the test antibody in its Fc domain, in the presence of a tumor-induced or -produced factors (TIPS factors). The amount of detectable label linked to the solid support in the presence of the TIPS factor is determined.

(17) In yet another aspect of the invention, a kit is provided for characterizing a test antibody comprising a Fab and a Fc domain. The kit comprises a first protein or aptamer that specifically binds to the Fab domain of the test antibody. The first protein or aptamer is attached to a solid support. The kit also comprises a second protein or aptamer that specifically binds to the Fc domain of the test antibody. The second protein or aptamer is labeled with a detectable agent. Alternatively, the first protein or aptamer binds to the Fc domain and the second protein or aptamer binds to the Fab domain.

(18) In one aspect of the invention a method is provided for identifying suitable pairs of protein or aptamer probes of antibody susceptibility to tumor-induced or -produced factors (TIPS factors). A pair of proteins or aptamers is tested for binding to an antibody and determining that both members of a pair can simultaneously bind. The pair of proteins or aptamers that are simultaneously bound to the antibody are tested for binding of the antibody to a cognate antigen of the antibody. Equivalent binding of the cognate antigen to the antibody in the presence and absence of the pair or protein or aptamers is determined.

(19) In an additional aspect of the invention, a composition is provided that comprises a first fluor-labeled protein or aptamer that specifically binds to the Fab domain of an antibody, and a second fluor-labeled protein or aptamer that specifically binds to the Fc domain of the antibody to form a dual-labeled antibody. The first fluor and the second fluor participate in fluorescence resonance energy transfer (FRET) when bound to the antibody. One protein or aptamer may be protein A and one may be protein L. Alternatively, one may be protein A and one may be the cognate antigen to which the antibody binds. The antibody may be present in the composition.

(20) In a further aspect of the invention a composition is provided that comprises an antibody, a first protein or aptamer that is bound to the antibody in its Fab domain, and a second protein or aptamer that is bound to the test antibody in its Fc domain. The first protein or aptamer is attached to a solid support. The second protein or aptamer is labeled with a detectable label. Alternatively, the first protein or aptamer is bound to the antibody in the Fc domain and the second protein or

aptamer is bound to the antibody its Fab domain.

(21) These and other aspects of the invention which will be apparent to those of skill in the art upon reading the specification provide the art with methods, compositions, and kits for use in improving anti-cancer antibody efficacy as well as anti-cancer therapeutic efficacy generally.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1. Schematic diagram of approximate positions of Fab/N-terminus and Fc/C-terminus domain paired probe sets to monitor antibody dynamic structure at steady-state and in the presence of TIPS factors. The figure indicates the approximate regions to which the antibody binding proteins Protein L (PL), Protein A (PA), Protein G (PG); the Fc Receptors (FcR or FcRn); the C1q complement initiating protein (C1q); the anti-Fab (α Fab) and anti-Fc (α Fc) aptamers; the anti-Fab (α Fab) and anti-Fc (α Fc) antibodies and antibody fragments; and the antibody-specific target antigen (antigen) bind, or to which fluors can be directly conjugated to amino acid (aa) residues within the Fab and/or Fc regions to generate a dual-labeled test antibody. The differing colors represent fluor 1 and fluor 2.

(2) FIGS. 2A-2B. Identifying complementary Fab and Fc pairs via ELISA using rituximab, a human-rodent chimeric antibody that can bind the CD20 antigen or a peptide containing the rituximab binding epitope (Reff M E, et al. Blood 83:435-445, 1994). Briefly, 96-well ELISA plates were coated with CD20 peptide (FIG. 2A), Protein L (FIG. 2B) or no protein or peptide and incubated with rituximab, a non-binding control human IgG1, or bovine serum albumin (BSA); and probed with biotinylated-Protein G, -human FCGR1A, or -human FCGR3A; and secondarily probed with STRP-HRP. The results show examples of the compatibility that probes can have for dual binding to a test antibody.

(3) FIG. 3. Effect of TIPS factors on dual-labeled test antibodies and probe selection. ELISA plates were coated with CD20 peptide (left set of bars), Protein L (middle set of bars) or BSA (right set of bars) and then probed with buffer (none), Protein G-biotin, FCGR1A-biotin or FCGR3A-biotin with or without the CA125 TIPS factor. Plates were washed and secondarily probed with streptavidin-HRP. As shown, the CA125 TIPS factor suppressed binding of the Fc domain probe FCGR3A-biotin in both the CD20 antigen and Protein L captured plate, while neither Protein G-biotin nor FCGR1A-biotin were affected, demonstrating and providing example of how the latter two sets are discovered to be appropriate for monitoring a dual-labeled test antibody's dynamic structure in the presence of a TIPS factor while FCGR3A is not.

(4) FIGS. 4A-4B. Effect of TIPS factor on a dual-labeled test antibody with complementary probes to the Fab and Fc domains labeled with a covalently linked anti-FabAF488 and an anti-FcAF555 antibody, respectively. Briefly, the dual-labeled test antibody is incubated with TIPS factors or control proteins and tested for effects on antibody dynamic structure via energy exchange between the complementary AF488 and AF555 fluors and captured via FRET. FIG. 4A shows a fluorogram of the dual-labeled test antibody employed. FIG. 4B shows the effect of TIPS factor CA125 or control HSA proteins at 10 ng/mL on a covalently linked dual-labeled test antibody or covalently linked single-labeled anti-FabAF488 and anti-FcAF488 test antibodies that are added together in presence of TIPS factors and control proteins. $P < 0.003$

(5) FIGS. 5A-5C. Effect of TIPS factors on a dual-labeled test antibody covalently linked to Protein L-AF555 (PL555) probe at the N-terminus and a covalently linked Protein A-AF488 (PA488) probe to the same test antibody's C-terminus. Briefly, the dual-labeled test antibody is incubated with TIPS factors or control proteins and tested for effects on antibody dynamic structure via energy exchange between complementary Alexa fluors AF488 and AF555 and captured via FRET. FIG. 5A shows a fluorogram (left) of the AF555-labeled Protein L and a denaturing SDS-PAGE gel with

unlabeled (left) or Alexa fluor labeled (right) Protein L (PL) and Protein A (PA) proteins. FIG. 5B shows a purified, dual-labeled test antibody covalently linked to PA488 and PL555 via NanoDrop One spectrophotometry/fluorometry with fluor peaks at 488 and 555 nm, respectively. FIG. 5C shows the effect of TIPS factor CA125 or control HSA proteins at 10 ng/ml on the covalently linked dual-labeled test antibody or covalently linked PL555 or PA488 single-labeled test antibodies alone or when added together in presence of TIPS factors or control proteins. These data demonstrate the ability of the dual-labeled test antibody to emit enhanced signal in the presence of TIPS factor but not control protein nor when test probe is single-label format. $P < 0.006$

(6) FIGS. 6A-6D. Effect of TIPS factor on dual-labeled test antibody with direct labeling of an amino acid residue with AF555 in the Fab domain and an AF488 fluor labeled Fc domain binding probe. Briefly, a dual-labeled test antibody is incubated with TIPS factors or control proteins and tested for effects on antibody dynamic structure via energy exchange between complementary fluors and captured via FRET. FIG. 6A is an ELISA demonstrating that a NaN.sub.3-treated RTX-169-DBCO-biotin labeled antibody was bound by STRP-HRP while a NaN.sub.3-treated RTX-WT-DBCO-biotin or untreated antibodies were not bound by STRP-HRP. A poly-lysine biotin-labeled RTX-169 (RTX-169-Bio full) served as a positive control for STRP-HRP binding. FIG. 6B demonstrates that the RTX-169-DBCO-biotin was labeled in the Fab domain where the unbound CYS 169 resides. Briefly, RTX-169-DBCO-biotin was papain digested, purified and the Fab and Fc domains were run on a non-denaturing SDS-PAGE gel and tested for STRP-HRP-biotin binding via western blot. As shown, STRP-HRP only bound the Fab fragment demonstrating the successful NaN.sub.3-mediated click chemistry conjugation of the DBCO-biotin to the Fab domain. Full length RTX-169-DBCO-biotin and RTX-169 unlabeled antibodies served as positive and negative controls, respectively. Staining of the membrane showed equal amounts of Fab and Fc fragments (not shown). MW represents the molecular weight marker in kDa. FIG. 6C shows a fluorogram of the dual-labeled RTX-169 test antibody employed, whereby AF555 is conjugated to the mutated CYS 169 residue within the N-terminal region of rituximab (SEQ ID: NO 26 and 27) and is dual-labeled through the conjugation of PA488 to the C-terminus. FIG. 6D shows the effect of TIPS factor CA125 or control HSA proteins at 10 ng/mL on the dual-labeled test antibody or test antibody single-labeled with AF555 and a separate test probe labeled with PA488 when added together in the presence of test proteins, whereby the TIPS factor stimulated emission of the dual-labeled test antibody but not controls. $P < 0.008$

DETAILED DESCRIPTION OF THE INVENTION

(7) The inventors have developed methods, reagents, and compositions that are useful in the development of improved antibodies for treating cancers and other diseases where humoral mediated therapies are available. While not wanting to be limited to any particular theory or mechanism of action, applicants believe that certain tumor-induced or -produced factors (TIPS) bind to antibodies and alter their structure such that they are no longer able to bind to their cognate antigen or engage with other elements of the humoral immune system. These disruptions may include suppression of the protein-antibody FcR-Ab binding (antibody-Fc receptors on immune effector cells, such as Natural Killer (NK), macrophage, monocytes, etc.) and/or C1q-Ab (classical antibody-complement complex) and subsequent ADCC and/or CDC (antibody-target antigen binding and/or alternative complement-mediated killing) activity; suppression of effector cell activation upon Ab-FcR engagement via SIGLECS (sialic acid binding Ig-like lectin) receptors; and suppression of alternative complement pathway recognition protein binding.

(8) In the assays taught and described below for measuring antibody binding to antigen or TIPS factors, any detectable label can be used. A fluorescent label is often convenient. Also useful are enzymatic, colorimetric, strong binding-pair, and radionuclear labels or agents. A detectable label may be immediately visible or detectable. Alternatively, it may require addition of other reagents to develop the detection. For example, a label can be an enzyme that requires particular substrates to make a product that is directly detected. Nonetheless, as referred to here, the enzyme itself is a

detectable label, because it can be readily detected by treatment with the particular substrate. Similarly, a member of a strong binding pair may be used that is actually detected only when its directly-labeled binding partner is added. Biotin-streptavidin is just one of many examples of such a strong binding pair.

(9) Some of the methods, compositions, and kits may be divided into three categories. In one category, an antibody, e.g., an anti-cancer antibody, is directly labeled, using two covalently attached fluors. One fluor is attached to the Fab or N-terminal domain and the other is attached to the Fc or C-terminal domain. The two fluors participate in FRET. In the second category, an antibody, e.g., an anti-cancer antibody, is bound by two proteins or aptamers that specifically bind to the antibody. The proteins or aptamers are labeled with a pair of fluors that participate in FRET. In the third category, one protein or aptamer is used and the antibody may have a unique amino acid engineered into its primary amino acid sequence that can be linked to a complementary fluor (see FIG. 6). In all categories, a change in FRET emission that occurs between the two fluors, whether to a protein or aptamer or attached directly to the antibody, indicates a change in the relationship of the two parts of the antibody molecule. Such a change can be used to detect and/or measure the effect of a TIPS factor on an anti-cancer antibody. Similarly, it can be used to measure the change in an antibody upon binding to its cognate antigen (also called its target antigen). A change in emissions can be determined as change in the intensity of emissions. Typically a change is determined when it is more than the amount that occurs using controls that do not cause a change in the antibody structure.

(10) Examples of proteins that may be used to bind to the Fc domain of a human antibody are Protein A, Protein G, human FCGR1A, FCGR2A, FCGR2B, FCGR2C, FCGR3A, FCGR3B, FCGR4, FCRL5, mouse FCGR4, and human C1q. A protein that may be used to bind to the Fab domain of an antibody is Protein L or the antibody's natural antigen. Antibodies raised against the test antibody's Fab or Fc domain are also examples of domain specific protein probes.

(11) Anti-cancer antibodies can be antibodies that are collected from a patient to be treated, collected from a different patient, or made in the laboratory by other means. Patient antibodies may be modified by various means prior to use. Certain commercial or experimental anti-cancer antibodies that may be analyzed using the methods of the invention include without limitation, rituximab, trastuzumab, trastuzumab emtansine, cetuximab, YP219, ocrelizumab, daratumumab, elotuzumab, alemtuzumab, necitumumab, pertuzumab, obinutuzumab, nivolumab, ipilimumab, pembrolizumab, ofatumumab, panitumumab, ibritumomab tiuxetan, sacituzumab govitecan, brentuximab vedotin and tositumomab.

(12) As shown in FIG. 3, a desirable a pair of proteins or aptamers that bind to an antibody desirably bind and remain bound to the antibody in the presence of antigen. Further, a desirable pair of proteins or aptamers also bind and remain bound to the antibody in the presence of a TIPS factor. This can be achieved with or without the use of crosslinking agents known by those skilled in the art.

(13) Once a dual-labeled antibody or a pair of proteins or aptamers that bind to an antibody is appropriately identified, they can be used to identify an antibody whose dynamic structure is altered by the presence of a TIPS factor. Such an antibody is termed a TIPS-Susceptible Antibody (TSA). With these components in place, one can test one or more substances, whether natural or synthetic, pure compounds or mixtures, or natural extracts, for example, to identify a substance that mitigates, ameliorates, or reverses the change in the antibody induced by the TIPS factor. Test substances may be found in collections or libraries that are commercially available, for example.

(14) Compositions of the invention can be formed in the course of conducting the methods. They may be pre-formed and packaged and provided to an entity that has a substance library to screen, for example. Similarly, the components of the assays and methods described here may be packaged together in a container and sold as a kit. The components of a kit need not be, but may be mixed together. They can be provided in separate containers or in a divided container, for example. Any

selection of antibodies, pairs of proteins or aptamers that bind to an antibody, labeled antibodies, TIPS factors, buffers, solid supports, and labels, described here may be formed as a composition or kit.

(15) While a few antibodies susceptible to one of potentially many TIPS factors are known, e.g., CA125, technologies that can identify these factors and TIPS Susceptible Antibodies (TSAs) whose dynamic structure is altered leading to suppressed humoral immune and pharmacologic activities to these effects are still needed.

(16) The “dynamic structure” of an antibody or protein is defined as the three-dimensional structures of an antibody at a given time point, wherein such time point coincides with said antibody's engagement with another protein or agent, and the structure of said antibody before this time point has changed into a different structure after this time point in response to the antibody's engagement with another protein or agent. The effect of a TIPS factor (protein or non-protein) binding to an antibody and affecting its dynamic structure has been reported in the case of hapten binding to the CDR domains of antibodies that in turn allosterically alter their Fc domain, thereby reducing its engagement with Fc Receptors (Janda A, et al. *Front Microbiol* 7:22, 2016). Previous studies also have shown that the dynamic structure of the (Fab').sub.2 domain is affected when an antibody was bound to its antigen (Werner T C, et al. *Proc Natl Acad Sci* 69:795-799, 1972). Recently, Kline et al reported that while several humanized monoclonal IgG.sub.1-type antibodies have similar amino acid structure, there are profound differences in the ability of a TIPS factor such as CA125 to bind them, suggesting either primary or secondary structure may dictate TIPS factor-antibody binding (Kline J B, et al. *OncoTarget* 8:52045-52060, 2017). In light of growing evidence that tumors utilize various pathways to avoid host immune defense and the fact that antibody-based approaches continue to be pursued against various cancer types, it is important to define methods to determine which antibodies are susceptible to TIPS factor binding that may result in structural dynamic change that can affect their engagement with humoral immune mechanisms and/or other functions as discussed here. These methods in turn enable the selection of lead antibodies from an antibody library based on resistance to TIPS factor suppressive effects and/or a means to identify said TIPS factor for patient screening. For example, if a test antibody is found to be affected by one or more TIPS factors, one may prescreen patients to avoid treating those that express a specific TIPS factor. Any preclinical-stage antibody may be used to screen patient samples for the presence of a specific TIPS factor as part of a drug development plan prior to full development, thereby saving the developers time in avoiding the costly investment required to develop an antibody that may be a TSA.

(17) We provide probes, kits and methods for screening an antibody's dynamic structure in the presence of known and unknown TIPS factors that can affect antibody dynamic structure and suppress its downstream immune-effector function(s) and/or cellular internalization upon binding to its target cell surface target antigen. The methods include a step of composing and employing matched or complementary probes that can simultaneously bind to the N-terminus or Fab domain and the C-terminus or Fc domain of a given test antibody, not affect the antibody's ability to bind to its natural target antigen and be formatted in a way that can determine a change in spatial distances of said probes in the presence of a TIPS factor derived from tumor membrane and/or as a soluble factor as compared to antibody alone. The probes (or “agents”) may comprise labeled antibody binding proteins such as Protein L and Protein A or G that are known to bind to the N- and C-termini of antibodies. Probes can also include labeled monoclonal antibodies (mAbs) and/or aptamers that specifically bind to the N- and C-termini of the test antibodies. Probes may also comprise a label-modified, antibody-specific antigen that when used in conjunction with a C-terminal probe (that may or may not be a natural Fc binding protein such as a Fc-gamma receptor or C1q) may define an antibody's dynamic structure alone or in the presence of a TIPS binding factor. These factors can be used in conjugated or non-conjugated formats to the test antibody. Finally, probes can be an antibody in which N-terminal and C-terminal residues are modified to

support site-specific labeling of a said antibody with acceptor-donor probes that can monitor the dynamic structure of the test antibody in the presence of soluble or membrane bound TIPS factors as shown in FIG. 6.

(18) FIG. 1 provides a schematic example of probes and their approximate binding regions. Probes comprising conjugated paired dyes or fluors may be used to measure the spatial distance of the N- and C-termini of a test antibody. These probes may be used in fluorescence resonance energy transfer (FRET) analysis whereby a fluorophore (donor) in an excited state from one probe transfers energy to a neighboring fluorophore (acceptor) of the complementary probe through dipole-dipole interaction, permitting one to measure the molecular proximity of the test antibody's N- and C-termini at Ångstrom distances. An increase or decrease of signal between test antibody alone versus test antibody plus TIPS factor suggest a binding and perturbation of the test antibody's dynamic structure and function.

(19) In one method for measuring the dynamic state of an antibody as a function of distance change between the test antibody's N- and C-termini, the antibody is first incubated with paired probes (labeled or unlabeled with a detectable agent) that can simultaneously bind to antibody, not affect the test antibody's binding to its antigen nor have a probe displaced by a TIPS factor upon its binding to test antibody. This can be achieved by any method capable of measuring probes in a sandwich format including ELISA. Such paired probes can then be labeled with detector agents (such as but not limited to fluors, enzymes, radionuclides, etc.) to effectively monitor the N- and C-terminal distance of said test antibody in solution alone as compared to antibody in solution with soluble or membrane-bound TIPS. In some aspects, the probes comprise antibody-binding proteins or derivatives thereof that include: (a) N-terminal (Fab domain) binding proteins or derivatives including Protein L (SEQ ID NO:1), a Fab specific antibody or antibody fragment, or the test antibody's specific antigen; (b) C-terminal (Fc domain) binding proteins or derivatives including Protein A (SEQ ID NO: 2), Protein G (SEQ ID NO: 3), a Fc specific antibody or antibody fragment; (c) Immunoglobulin Fc receptors (bind the Fc domain) of human FCGR1A (SEQ ID NO: 4), FCGR2A (SEQ ID NO:5), FCGR2B (SEQ ID NO:6), FCGR2C (SEQ ID NO:7), FCGR3A (SEQ ID NO:8), FCGR3B (SEQ ID NO:9), FCGRT (SEQ ID NO:10), FCRL5 (SEQ ID NO:11), mouse FCGR4 (SEQ ID NO: 12), human C1q (SEQ ID NO:13) and/or derivatives thereof.

(20) The paired probes may comprise aptamer pairs that can specifically and simultaneously bind to the N- and C-termini of an antibody and do not affect test antibody binding to its antigen where said aptamers are labeled with a donor and acceptor detector agent (i.e., fluorophore (fluor), enzyme, radionuclide, etc) and tested for N- and C-termini proximity when bound to a test antibody alone or in the presence of potential soluble or membrane TIPS factors. TIPS factors that bind and "affect" the spatial distance of the complementary probes are identified as potential factors that impose suppression of humoral response(s) including pharmacokinetic (PK), pharmacodynamic (PD) and pharmacologic (PL) suppression, including cellular internalization or antigen binding of test antibody. The term "affect" generally refers to a 8% or greater change in signal of paired probes when incubated with test antibody alone as compared to antibody with TIPS factor. It may, depending on the antibody and the probes used also refer to a change of at least 5%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 70%, or 75%.

(21) In some aspects of the compositions and methods, donor and acceptor detector agents linked to paired probes comprise Alexa fluors (AF):

(22) TABLE-US-00001 Absorption Emission Emission Alexa Fluor Dye Max (nm) Max (nm)
Color Alexa Fluor 350 346 442 Blue Alexa Fluor 405 402 421 Blue Alexa Fluor 430 434 539
Yellow-green Alexa Fluor 488 495 519 Green Alexa Fluor 514 518 540 Green Alexa Fluor 532 531
554 Yellow Alexa Fluor 546 556 573 Orange Alexa Fluor 555 555 565 Orange Alexa Fluor 568
578 603 Red-orange Alexa Fluor 594 590 617 Red Alexa Fluor 610 612 628 Red Alexa Fluor 633
632 647 Far-red Alexa Fluor 635 633 647 Far-red Alexa Fluor 647 650 668 Far-red Alexa Fluor
660 663 690 Near-IR § Alexa Fluor 680 679 702 Near-IR § Alexa Fluor 700 702 723 Near-IR §

Alexa Fluor 750 749 775 Near-IR § Alexa Fluor 790 782 805 Near-IR §

(23) Yet in other aspects of the methods donor and acceptor detector agents comprise fluors:

(24) TABLE-US-00002 Absorption/ Fluor Emission (nm) Coumarin (AMCA) 346/442 Cy2 or Fluorescein 495/519 (FITC) Cy3 or 555/565 Tetramethylrhodamine (TRITC) Rhodamine Red 587/603 Texas Red 590/617 Cy5 650/668 Cy5.5 679/702 Cy7 749/775

(25) Methods of screening for paired probes to test antibodies comprising Fab and Fc domains may comprise testing probes for: (1) ability to bind test antibody and not affect said antibody's ability to bind its target antigen; and (2) ability of both probes to stay bound by test antibody in the presence of known candidate or unknown TIPS factors. Probe pairs meeting these specifications may then be labeled with detector agents that can measure antibody dynamic structure in the presence of known or unknown TIPS.

(26) Bispecific (BSP) antibodies may be screened for those that may be affected by soluble or membrane bound TIPS factors that may alter their PK, PD and/or PL activity, including inability to bind one or both of test antibody's target antigen(s). The BSP may be probed with N- and C-terminal paired probes and tested for dynamic structure in the presence or absence of soluble or membrane-bound TIPS factors. The BSP test antibody-specific antigens may be labeled with acceptor/donor detector agents (i.e., fluors, enzymes, radionuclides, etc.) and the dynamic structure may be tested to antigen bound BSP in the presence or absence of soluble or membrane TIPS. Those “affected” may then be tested for pharmacokinetic (PK), pharmacodynamics (PD) and pharmacological (PL) activity in presence of binding TIPS factors.

(27) Methods for screening antibody drug conjugates (ADCs) that may be affected by soluble or membrane bound TIPS factors that may alter their PK, PD or PL activity, include cellular internalization. The ADC may be probed with N- and C-terminal probes and tested for dynamic structure in the presence or absence of soluble or membrane-bound TIPS. Those “affected” may then be tested for PK, PD and PL activity, including cellular internalization in presence of binding TIPS factors.

(28) Labeled paired probes may be added to the test antibody in the presence of a crosslinking agent.

(29) Protein L may be conjugated to the Alexa fluor AF555 and used in combination with Protein A conjugated to AF488.

(30) An engineered rituximab (SEQ ID NO: 26 and 27) may be generated to have an unpaired cysteine in its Fab domain and used to directly conjugate a fluor, which includes but is not limited to AF555. The engineered antibody may then be then conjugated to a Protein A labeled with AF488 and used to monitor the binding of TIPS factor to test antibody.

(31) Various terms and terminology (“terms”) relating to aspects of the enclosed description are used throughout the specification and claims of this document. Such terms are to be given their ordinary meaning in the art unless otherwise specifically indicated. Other specifically defined terms are to be construed in a manner consistent with the definitions provided.

(32) As used in this specification and the appended claims, the singular forms of “a,” “an,” and “the” also include plural references unless the content clearly specifically dictates otherwise. Reference to “a probe” may include combination of a pair of complementary probes capable of measuring the distance between two probes via any analytical method known by those skilled in the art. Similarly, reference to “a cell” may include a combination of two or more cells, and the like.

(33) The term “about” as used when referring to a quantified values such as an amount, a period of time, molecular distance, and/or the like, is meant to encompass variations of up to $\pm 8\%$ from the specified value, as such variations are appropriate to carry out the disclosed methods. Unless otherwise indicated, all values expressing quantities of reagents, such as molecular weight, molarity, reaction conditions, molecular distance and so forth used in the specification and claims are to be understood as being quantified in all instances by the term “about.” Accordingly, unless

indicated to the contrary, the numerical values as set forth in the following specifications and listed claims are approximations that may vary depending upon the desired properties of the composition agent and/or methods sought to be obtained by the present invention. At the very least, each numerical value should at least be valued by the reported significant digits and through ordinary rounding methods known by those skilled in the art.

(34) The term “antibody” (also referred to as “Ab”) as used is meant in a broad sense and includes immunoglobulin (also referenced as “Ig”) or antibody molecules including polyclonal antibodies (also referenced as pAbs), monoclonal antibodies (also referenced as mAbs) including murine, rat, monkey, human, humanized and chimerized mAbs, bispecific antibodies (also referenced as BSP), antibody drug conjugates (also referenced as ADCs), antibody fused immunotoxins and antibody fragments. In general, antibodies are proteins or polypeptide chains that bind to a specific antigen. An antigen is a structure that is specifically recognized by a given antibody. Canonical antibodies comprise heterotetramer glycosylated proteins, composed of two light chains and two heavy chains lined through a complex of disulfide and hydrogen bonds. Each heavy chain has a variable domain (variable region) (VH) followed by a number of constant domains (referred to as the Fc domain). Each light chain has a variable domain (VL) as well as a constant domain; the constant domain of the light chain is aligned with the first constant domain of the heavy chain and the light chain VL is aligned with the variable domain of the heavy chain. Antibody light chains of any species are assigned to one of two distinct types based on their amino acid sequences within their constant domains, namely kappa (κ) and lambda (λ). For compositions included herein, reference to light chain involves either subtype.

(35) Immunoglobulins are categorized as classes or isotypes, depending upon the type of Fc domain namely IgA, IgD, IgE, IgG and IgM, which depend on the sequences contained within their heavy chain constant domain. The IgA and IgG isotypes are further comprised of subclasses as the isotypes IgA.sub.1, IgA.sub.2, IgG.sub.1, IgG.sub.2, IgG.sub.3 and IgG.sub.4. For compositions included herein, reference to heavy chain involves any subtype.

(36) An immunoglobulin light or heavy chain variable region consists of a “framework” region interrupted by three Complementarity Determining Regions (CDRs) also referred to as “antigen-binding sites” based on sequence variability as reported (Wu T T, Kabat E A. J Exp Med 132:211-250, 1970). In general, an antigen-binding site is composed of six CDRs with three located within the variable heavy chain (CDRH1, CDRH2, CDRH3), and three within the variable light chain (CDRL1, CDRL2, CDRL3) (Kabat E A, et al. 5.sup.th Ed. PHS, National Institutes of Health, Bethesda, Md., 1991).

(37) Antigen-binding or antibody binding fragments are any structure that may exhibit binding affinity for a particular antigen. Some fragments are composed of portions of antibodies that retain antigen-binding specificity of the parent antibody molecule. In some instances, fragments may comprise at least one variable region (either a heavy chain or light chain variable region) or one or more CDRs of an antibody known to bind to a particular antigen that may be complexed with a variable region of another antigen, remain in single chain format or formatted as a peptide to retain binding to target antigen. Examples of fragments include, without limitation bispecific, diabodies and single-chain molecules as well as Fab, (Fab').sub.2, Fc, and single chain (ScFv) antibodies, individual antibody light chains, individual antibody heavy chains, chimeric fusions between antibody chains or CDRs and other proteins, protein scaffolds, heavy chain monomers or dimers, light chain monomers or dimers, dimers consisting of one heavy and one light chain, and the like. Any antibody isotype may be used to produce an antibody or antigen-binding fragments.

Additionally, fragments may include non-antibody proteins consisting of frameworks that may successfully incorporate polypeptide segments in an orientation that confers affinity for a given antigen of interest, such as protein scaffolds derived from adhesion-type proteins including the fibronectins and collagens. Antigen-binding fragments may comprise non-protein scaffolds such as aptamers. Aptamers may comprise nucleic acids, such as RNA, DNA, or nucleic acids with non-

classical nucleotides, for example. The phrase “an antigen-binding fragment thereof” may be used to denote that a given antigen-binding fragment incorporates one or more antigen-binding segments of the antibody binding fragment referred to in the phrase. An antibody-fragment may also be referred to as a “probe.”

(38) “Specific binding” or “specifically binds” refers to the binding of an antibody, antibody fragment, antigen-binding fragment or aptamer to an antigen (including sequences contained within an antibody itself) with greater affinity than for other antigens. Typically, a specific antibody, antigen-binding fragment or aptamer binds target antigen with an equilibrium dissociation constant $K_{\text{sub.D}}$ of about 9×10^{-8} M or less.

(39) An “antibody derivative” means an antibody, as defined above, that is modified by covalent attachment of another molecule such via peptide chemistry (i.e., amidation, etc.), chemical conjugation, genetic fusion and/or via post translational moieties (i.e., glycosyl, acetyl and/or phosphoryl) not typically associated with the antibody, and the like.

(40) The term “antibody dynamic structure” refers to any change in structure that can affect the signal of two probes (i.e., antibody binding proteins (Protein L, A, G, etc; aptamers; natural mammalian binding proteins (i.e., Fc receptors, C1q, etc); Fab and Fc binding antibodies; haptens; and/or antibody specific antigens) that are attached to the antibody.

(41) The term “monoclonal antibody (mAb)” refers to an antibody that is derived from a single cell clone, including any eukaryotic or prokaryotic cell clone, or a phage clone, and not the method by which it is produced. Thus, the term “monoclonal antibody” is not limited to antibodies produced through hybridoma technology but may also include recombinant methods.

(42) “Fab domain” refers to any antibody sequence N-terminal to the antibody hinge disulfide region which is known by those skilled in the art.

(43) “Fc domain” refers to any antibody sequence C-terminal to the antibody hinge disulfide region which is known by those skilled in the art.

(44) An “antigen” is an entity to which an antibody, antibody fragment or aptamer specifically binds. This includes binding to a said antibody or protein of interest.

(45) The term “tumor-induced or -produced factor (TIPS or TIPS factor)” refers to any protein or non-protein factor that is directly produced by a dysregulated cell or induced from normal cells by a dysregulated malignant cell. It has been reported that the CA125 protein is produced by dysregulated cells such as ovarian carcinoma and mesothelioma (Kline J B, et al. Eur J Immunol 48:1872-1882, 2018), as well as induced by lymphomas from normal surrounding epithelial cells (Sanusi et al. Perit Dial Int. 21:495-500, 2001).

(46) The term “dysregulated cell” refers to any cell that behaves of functions differently than its normal counterpart. This includes transformed cells by pathogens as well as malignant cells derived from tumors.

(47) The term “TIPS-inhibitor” refers to any protein or non-protein agent that can block a tumor-induced or -produced factor and/or a factor from a dysregulated cell from binding to test antibody.

(48) The term “test antibody” refers to an antibody in which two probes have been attached, one to the Fab domain and the other to the Fc domain and screened to determine if test antibody is bound by TIPS factor and has an effect on its dynamic structure.

(49) The term “CA125” refers to the TIPS gene product produced by MUC16 gene (HGNC: 15582; OMIM: 606154), which is found in soluble and membrane-bound forms. It has been reported to bind to antibodies and affect bound antibody humoral immune function (Kline J B, et al. Oncotarget 8:52045-52060, 2017).

(50) The term “unknown TIPS” refer to TIPS that are not known to bind to antibody, BSP or ADC or whose composition is unknown at the time of screening.

(51) The terms “cancer,” “malignant,” and “tumor” are well known in the art and refer to the presence of cells with -dysregulated cell growth and morphological features different than a normal cell type of similar origin. Malignant refers to those cancer cells capable of causing morbidity

and/or mortality. As used, “cancer and tumor” includes premalignant and malignant types.

(52) As used, the term “soluble” refers to a protein or non-protein agent that is not attached to the cellular membrane of a cell. For example, an agent that is soluble may be shed, secreted or exported from normal, dysregulated or cancerous cells into biological fluids including serum, whole blood, plasma, urine or microfluids of a tumor.

(53) The term “labeled,” with regard to a test antibody's binding agent (also referred to as a “probe”), is intended to encompass direct labeling of the probe by coupling (i.e., physically linking) a detectable substance to the probe (such as but not limited to a fluor, enzyme, radionuclide, etc.), as well as by indirect labeling of the probe by reactivity with another reagent that is directly labeled. An example of indirect labeling includes detection of a primary probe via a secondary fluorescently labeled probe that may include an antibody or aptamer that is specific to said probe. The primary or secondary probe may be labeled via radionuclides, chromophores, fluorophores, or enzymes. The probe or secondary probe may be an antibody derivative, an antibody fragment, a protein scaffold capable of target specific binding or an aptamer that is labeled with a protein ligand (e.g., biotin, the ligand for avidin and streptavidin).

(54) The term “detector agent” or “detectable agent” refers to any agent that can be linked to a protein, a protein binding agent including aptamers and detected via devices commonly used in the field. These include but are not limited to fluors, enzymes and enzyme substrates, radionuclides, heavy metals and colorimetric dyes and substrates that can be detected via methods such as but not limited to densitometry, spectrophotometry, luminescence, microscopy, radiography and scintillation. Additional reagents may be needed to develop the detectable signal. The detector agent may be an alexa fluor AF555 covalently labeled Protein L.

(55) The term “complementary fluor” refers to two fluorescent moieties in which one can act as a donor upon excitation at one wavelength to the second fluor agent that acts as an acceptor and emits specific wavelength signal and intensity that is proportional to the absolute distance of the two fluor agents and is commonly used in FRET-based assays by those skilled in the art. This can be any combination of donor and acceptor fluors.

(56) The “level” of a specified protein or non-protein agent, as used, refers to the level or levels of the agent as determined using any method known in the art for the measurement of protein and/or non-protein agent levels in vitro or in vivo. Such methods include gel electrophoresis, capillary electrophoresis, high performance liquid chromatography (HPLC), thin layer chromatography (TLC), hyperdiffusion chromatography, fluid or gel precipitation reactions, absorption spectroscopy, colorimetric assays, spectrophotometric assays, flow cytometry, immunodiffusion (single or double), solution phase assay, immunoelectrophoresis, Western blotting, radioimmunoassay (RIA), enzyme-linked immunosorbent assays (ELISAs), immunofluorescent assays, fluorescence resonance energy transfer (FRET), Förster resonance energy transfer, electrochemiluminescence immunoassay, and the like. The level of an TIPS factor (e.g., CA125) may be determined using probe-based techniques, as described in more detail.

(57) The term “dual-labeled test antibody” refers to one in which a fluor is attached directly or indirectly to a residue(s) in the Fab domain and a complementary fluor to residues in the Fc domain.

(58) The term “TIPS susceptible antibody (TSA)” refers to any antibody, antibody fragment, BSP or ADC that is directly bound by a TIPS factor whose dynamic structure is altered. It has been reported that the CA125 TIPS factor, which is produced by malignant cells such as ovarian carcinoma and mesothelioma (Nicolaides N C, et al. Cancer Biol Ther 19:622-630, 2018) as well as induced by lymphomas from normal surrounding epithelial cells (Sanusi et al. Perit Dial Int. 21:495-500, 2001), can bind certain antibodies and alter their dynamic structure thus affecting their biological activities including ADCC, CDC, opsonization, internalization and/or PK profiles.

(59) The term “antibody drug conjugate (ADC)” refers to any antibody that is conjugated or fused to a chemical or polypeptide that has toxic activity to cells.

- (60) The term “bispecific antibody (BSP)” refers to any antibody that can bind two or more different antigens. A BSP can comprise at least but not limited to two full length antibodies, a full length antibody and a single chain antibody, or two single chain antibodies, whereby one each binds to different antigens or different epitopes on the same antigen.
- (61) The term “antibody dependent cellular cytotoxicity (ADCC)” refers to a process where an antibody can bind to an antigen on a surface of a cell then engage with immune cells via sequences within said antibody's Fc domain that in turn results in immune cells releasing toxins that can kill bound cell.
- (62) The term “complement dependent cytotoxicity (CDC)” refers to a process where an antibody can bind to an antigen on a surface of a cell then engage with the C1q protein via sequences within said antibody's Fc domain that in turn results in initiation of classical complement cascade that can kill bound cell.
- (63) The term “internalization” refers to a process where an antibody, antibody fragment or ADC can bind to an antigen on a surface of a cell then internalize via mechanisms known to those skilled in the art.
- (64) The term “pharmacokinetic (PK)” refers to the time that an antibody maintains its steady state concentration when administered to a subject.
- (65) The term “pharmacodynamic (PD)” refers to the study of the biochemical and physiological effects of an antibody-based drug and its mechanisms of action(s), including the correlation of their actions and effects with their biochemical structure when administered to a subject.
- (66) The term “pharmacologic (PL)” refers to the known effect an antibody has on a managing or killing a disease cell in vitro or in vivo.
- (67) The term “sample” refers to a collection of similar fluids, cells or tissues isolated from a subject, as well as fluids, cells or tissues present within a subject. Fluids may include biological fluids that include liquid solutions contacted with a subject or biological source, including cell and organoid culture medium, urine, salivary, lavage fluids and the like.
- (68) The term “control sample,” as used, refers to any biologically or clinically relevant control sample, including, for example, a sample from a healthy subject not afflicted with a particular cancer type or a protein known not to bind test antibody.
- (69) The term “control level” refers to an accepted or pre-determined level of a protein or non-protein agent that is used to compare with the level of the same agent in a sample derived from a subject or is the baseline of non-specific protein binding to test antibody.
- (70) As used, “a difference” between signal of an antibody in control vs being bound by a TIPS agent is generally any difference that can be statistically determined using statistical methods commonly used by those skilled in the art and at a minimum a difference of 8% or greater as compared to control. It may, depending on the antibody and the probes used also refer to a change of at least 5%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 70%, or 75%.
- (71) The term “inhibit” or “inhibition of” means to reduce by a statistically measurable amount, or to prevent entirely.
- (72) The term “functional,” in the context of an antibody, antibody fragment or antibody containing moiety (i.e. BSP, ADC, etc.) to be used in accordance with the methods described, indicates that the antibody is capable of binding to antigen and/or is able to bind to and kill target cell in vitro or in vivo.
- (73) The term “target cell” refers to a cell or population of cells that expresses antigen for a specific antibody or antibody containing moiety. These can be derived directly from patients or be from cell lines.
- (74) The term “effector cell” refers to any cell that can bind to an antibody and induce a killing effect on target cell. These include but are not limited to NK cells, CD3+ T-cells, CD8+ T cells, monocytes, macrophages, dendritic cells etc.
- (75) The term “pharmaceutically acceptable” refers to a substance that is acceptable to administer

to a patient from a pharmacological as well as toxicological aspect and is manufactured using approaches known by those skilled in the art. These include agents approved by a regulatory agency of the Federal or a state government or listed in the U.S. Pharmacopeia or other generally recognized pharmacopeia for use in animals and humans. The term “pharmaceutically compatible ingredient” refers to a pharmaceutically acceptable diluent, adjuvant, excipient or matrix vehicle with which an anti-cancer agent is administered. “Pharmaceutically acceptable carrier” refers to a matrix that does not interfere with the effectiveness of the biological activity of the active ingredient(s) and is nontoxic to the host.

(76) The terms “effective amount” and “therapeutically effective” are used interchangeably and, in the context of administering a pharmaceutical agent at an amount that is sufficient to produce an enhanced clinical outcome in a patient. An effective amount of an agent is administered according to the methods described here in an “effective regimen.” The term “effective regimen” refers to a combination of amount of the agent and dosage frequency adequate to accomplish an enhanced clinical outcome for a patient with a particular cancer who may be selected for a particular TIPS profile. Enhanced efficacy is an improved clinical outcome when a patient is administered an agent that is capable of overcoming morbidity better than a said parental compound or an agent that can enhance the clinical outcome of an effective regimen.

(77) The terms “patient” and “subject” are used interchangeably to refer to humans and other non-human animals, including veterinary subjects, that receive a therapeutic agent treatment. The term “non-human animal or non-human” includes all vertebrates. In a preferred embodiment, the subject is a human.

(78) The term “non-human” refers to all vertebrates excluding *Homo sapiens*.

(79) “Therapeutic agents” are typically substantially free from undesired contaminants. This means that an agent is typically at least about 50% w/w (weight/weight) pure as well as substantially free from interfering proteins and contaminants.

(80) The term “screening” may refer to testing of TIPS factors that can bind to a test antibody, antibody fragment or antibody containing moiety (i.e., BSP, ADC, single chain antibody, antibody fragment, etc.) using, e.g., two probes to the antibody that can monitor the distance between the two when bound to antibody, antibody fragment or antibody containing moiety when alone and when combined with potential TIPS or unidentified TIPS factors within fluids from patients or subjects. The term may be used in other contexts in which a large number of test elements is being assayed to determine which among the test elements has a certain property. Similarly it can be used to refer to the assaying of patient samples for those having a particular property.

(81) Composition of Probes, Kits and Methods for Monitoring Antibody Dynamic Structure in the Presence of Tumor-Induced and/or -Produced Factors (TIPS)

(82) Compositions of probes, kits and methods may be used for identifying susceptible antibodies to TIPS factors that may bind a test antibody and affect its dynamic structure [here referred to as TIPS Susceptible Antibody (TSA)] as well as for identifying tumor-induced or -produced factors (here referred to as TIPS) that may affect antibody dynamic structure, as well as methods of using probes and kits for detecting TSAs and/or known or unknown TIPS factors capable of binding antibodies. The method may involve the binding of two probes where one binds to the Fab domain and the other the Fc domain. The Fab binding probe is an aptamer specific to antibody N-terminal sequences, Protein L, a Fab-specific antibody, the test antibody's specific antigen, or a small chemical that can specifically conjugate to a natural or non-natural residue(s) in the Fab domain. The Fc binding probe may be an aptamer specific to C-terminal sequences, Protein A, Protein G, a Fc-specific antibody, a natural mammalian Fc binding protein (i.e. Fc receptors, FcRn, C1q, etc), or a small chemical or linker that can specifically conjugate to a natural or non-natural residue(s) in the Fc domain. Examples are schematically shown in FIG. 1 and experimentally shown in FIGS. 4-6. Kits may be composed of said probes that are able to detect a change in distance between N-terminal and C-terminal probes when a TSA is affected by a TIPS factor. Distance may be

measured in angstroms and can be detected by any instrument capable of measuring signal of probes with detection labels used by those skilled in the art.

(83) In the methods for identifying a TSA, a test antibody is added to a complementary set of Fab and Fc domain probes in presence of a candidate TIPS factor or unknown TIPS factors and signal is compared to dual-labeled test antibody plus probes alone to determine if candidate TIPS factors may change antibody dynamic structure by bringing probes closer or farther apart, thereby altering the signal produced between probes. Complementary probes refer to those that can be used in combination to detect a signal. For example, a Fab domain probe may be labeled with a fluor that emits energy when exposed to certain light wavelengths that in turn stimulate the signal of the Fc domain probe that contains a fluor capable of stimulated emission by the fluor contained Fab probe. These signals (wavelengths, enzymatic activity, temperature, etc.) can be captured by any machine or device that can detect a fluorescent, colorimetric, thermographic, luminescent or visual change in signal. If the test antibody is bound by a TIPS factor that results in altered antibody structure, the wavelength between two complementary probes will change thus signaling that the antibody is a TSA. A change of at least 8% is typically considered as being a meaningful effect on dynamic structure as it can perturb TSA ADCC and CDC functions. Depending on the antibody and the probes used a meaningful effect also may be defined as a change of at least 5%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 70%, or 75%.

(84) For TIPS factor and TSA assessments, the sample tested for a known or unknown TIPS factor binding and subsequent TSA activity may be a purified protein or non-protein factor produced recombinantly, synthetically or derived from natural sources. Natural sources include whole blood, serum, plasma, pleural effusions, ascites, tumor tissues from surgical resection or biopsies, histological preparations, and the like. Alternatively, a protein fraction can be isolated from a dysregulated or malignant cell's soluble or membrane compartments and analyzed for TIPS factor binding and TSA effect on test antibody. For example, the step of determining if a TIPS factor has test antibody binding and TSA activity may involve labeling test antibody with N- and C-terminal complementary fluor probes, then determine its signal using FRET, comparing signal when antibody is alone or after mixture with a known or unknown TIPS factor. Generally, the antibody is either directly labeled with specific complementary dyes or fluors capable of producing a detectable signal as a dual labeled test antibody or bound by complementary N- and C-terminal probes. Examples of types of labels include enzyme labels, radioisotopic labels, nonradioactive labels, fluorescent labels, toxin labels and chemoluminescent labels. Many such labels are readily known to those skilled in the art. For example, suitable labels include, but are not limited to radiolabels, fluorescent labels [such as Alexa fluor 488 (AF488) or 555 (AF555)], epitope tags, biotin, chromophore labels, ECL labels, or enzymes. More specifically, the described labels include ruthenium, .sup.111In-DOTA, .sup.111In-diethylenetriaminepentaacetic acid (DTPA), horseradish peroxidase, alkaline phosphatase and beta-galactosidase, poly-histidine (HIS tag), acridine dyes, cyanine dyes, fluorine dyes, oxazin dyes, phenanthridine dyes, rhodamine dyes, Alexa Fluor® dyes, and the like. Detection of a signal from the dual-labeled test antibody indicates that the TIPS factor has TSA activity, which is typically indicated by a signal change of 8% or greater between conditions. Depending on the antibody and the probes used the signal change may be at least 5%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 70%, or 75%.

(85) The level of a TIPS factor with TSA activity in a biological sample can (but need not) be determined with respect to one or more standards. The standards can be historically or contemporaneously determined. The standard can be, for example, a biological sample known not to be cancerous from a different subject. The standard can also be the patient sample under analysis contacted with an irrelevant antibody known to be resistant to TSA activity.

(86) The use of TIPS factor that can produce a TSA effect on the test antibody can also be formatted to screen for inhibitors that may bind a particular TIPS factor. The TSA may be labeled to produce a dual-labeled test antibody or with a complementary set of probes to the Fab (N-

terminal) and Fc (C-terminal) domains and a TIPS protein is added to labeled TSA in the presence of a library of compounds and screened for inhibition of TIPS binding to TSA. A dual-labeled test antibody may be used to screen a compound library in the presence of a known TSA, where an inhibitory compound may block the signal of said TSA to test antibody. The library consists of small chemical compounds. The library may consist of peptides and/or proteins including antibodies and antibody fragments. The library may also comprise natural products, such as human or non-human-derived serum proteins, soil samples or plant extracts. Compounds found to block TIPS factor ability to bind TSAs via FRET or other assays are referred to as TIPS factor inhibitors.

(87) In screening for TIPS factor inhibitors, a hybrid format may be used in which one of the detectable labels that participates in FRET is attached to an amino acid residue that is part of the antibody, in the Fab or the Fc domain, and one of the detectable labels that participates in FRET is attached to a probe that binds to the Fab or Fc domain. Typically, the one of the two detectable labels is attached to the Fab domain and one to the Fc domain. When testing FRET of two labels, irrespective of how they are attached or bound to the domains of the antibody, one can test in the presence and absence of TIPS factor. The TIPS factor may bind to the antibody and change the FRET. When a potential TIPS factor inhibitor is contacted with the dual labeled antibody bound to a TIPS factor, FRET can again be measured. This measurement may be done in the presence and in the absence of the potential TIPS factor inhibitor. Similarly, the measurement may be done on the dual labeled antibody bound to a TIPS factor which has been contacted with the potential TIPS factor inhibitor and on the dual labeled antibody bound to a TIPS factor that has not been contacted with the TIPS factor inhibitor.

(88) A cancer subject may be treated with a TIPS factor inhibitor. For example, a patient may have a mesothelin-expressing cancer such as colorectal, lung, ovarian, pancreatic, endometrial carcinoma or mesothelioma. Several anti-mesothelin antibodies have been reported to bind to TIPS factor, making a TIPS factor inhibitor a desirable entity.

(89) A patient with mesothelin-expressing cancer that expresses a TIPS factor may be treated with a TIPS factor inhibitor. An anti-mesothelin therapeutic agent is administered to a subject along with the TIPS-inhibitor. The TIPS factor may be CA125. An anti-mesothelin therapeutic agent may be administered to the subject, where the subject has a baseline CA125 level that is above the normal range. The TIPS factor inhibitor may be administered alone. The TIPS factor inhibitor may be administered in combination with chemotherapy. The chemotherapy may be cisplatin, carboplatin and/or pemetrexed, or any other chemotherapeutic agents considered standard of care at the time when the subject is treated. CA125 expression level may be determined by any means known in the art.

(90) An anti-mesothelin therapeutic agent may be an antibody that specifically binds to mesothelin, preferably to mesothelin expressed on mesothelioma, lung adenocarcinoma, or colorectal cells. Alternatively, antigen-binding fragments of such an antibody, derivatives, and variants may be used for treatment. An exemplary antibody that specifically binds to mesothelin may be an antibody selected from the group consisting of: (a) an antibody comprising YP219 antibody CDRs: SEQ ID NO:14 (GFDLGFYFYAC) as CDRH1, SEQ ID NO:15 (CIYTAGSGSTYYASWAKG) as CDRH2, SEQ ID NO:16 (ARSTANTRSTYYLNL) as CDRH3, SEQ ID NO:17 (QASQRISSYLS) as CDRL1, SEQ ID NO:18 (GASTLAS) as CDRL2 and SEQ ID NO: 19 (QSYAYFDSNNWHA) as CDRL3, numbered according to Kabat; (b) an antibody that binds the same epitope as YP218 (Zhang et al. Scientific Reports volume 5, Article number: 9928, 2015).

(91) An antibody that specifically binds to mesothelin may comprise a mature light chain variable region comprising the amino acid sequences of SEQ IDS NO: 17, 18 and 19 and/or a mature heavy chain variable region comprising the amino acid sequences of SEQ IDS NO: 14, 15 and 16. The anti-mesothelin therapeutic agent may be YP219. Other useful antibodies that specifically bind to mesothelin comprise mature light and heavy chain variable regions having at least 90% and preferably at least 95%, 96%, 97%, 98% or 99% sequence identity to SEQ IDS NO: 17, 18 and 19,

or SEQ IDS NO: 14, 15 and 16. Other useful anti-mesothelin antibodies or derivatives thereof can competitively inhibit binding of YP219 to mesothelin, as determined, for example, by immunoassay. A derivative of an anti-mesothelin antibody may also be used in the practice of present methods. Typical modifications to make such derivatives include, e.g., glycosylation, deglycosylation, acetylation, pegylation, phosphorylation, amidation, derivatization by known protecting/blocking groups, proteolytic cleavage, linkage to a cellular ligand or other protein, cytotoxic small molecule and the like. Additionally, the derivative may contain one or more non-classical amino acids.

(92) The present methods can be combined with other means of treatment such as surgery (e.g., debulking surgery), radiation, targeted therapy, chemotherapy, immunotherapy, use of growth factor inhibitors, or anti-angiogenesis factors. An anti-mesothelin therapeutic agent along with a TIPS factor inhibitor can be administered concurrently to a patient undergoing surgery, chemotherapy or radiation therapy treatments. Alternatively, a patient can undergo surgery, chemotherapy or radiation therapy prior to or subsequent to administration of the anti-mesothelin therapeutic agent and TIPS factor inhibitor by at least an hour and up to several months, for example at least an hour, five hours, 12 hours, a day, a week, a month, or three months, prior or subsequent to administration of the anti-mesothelin therapeutic agent. Administration of a therapeutically effective amount of a platinum-based chemotherapy and/or a folate antimetabolite may be given to the subject in addition to the anti-mesothelin therapeutic and TIPS factor inhibitor agents. For example, administration of therapeutically effective amounts of an anti-mesothelin antibody, a TIPS factor inhibitor, a platinum-based chemotherapy, and/or a folate antimetabolite may be administered. The platinum-based chemotherapy may be cisplatin, carboplatin, oxaliplatin, satraplatin, picoplatin, nedaplatin, triplatin, lipoplatin, or combinations thereof. The platinum-based chemotherapy may be any other platinum-based chemotherapy known in the art. The folate antimetabolite is pemetrexed. The platinum-based chemotherapy may be administered to the subject once every week, once every two weeks, once every three weeks, or once every four weeks. The folate antimetabolite may be administered to the subject once every week, once every two weeks, once every three weeks, or once every four weeks. When both a folate antimetabolite and a platinum-based chemotherapy are administered to the subject as part of the treatment regimen, the anti-mesothelin therapeutic agent, the TIPS factor inhibitor, the platinum-based chemotherapy, and the folate antimetabolite may be administered sequentially in any order or simultaneously.

(93) A subject may have received first-line surgical resection of the tumor, first-line platinum-based therapy, first-line folate antimetabolite-based therapy, and/or first-line platinum and folate antimetabolite-based therapy for treatment of a mesothelin-expressing cancer prior to administering anti-mesothelin antibody and TIPS factor inhibitor.

(94) Administration of the therapeutic agents (including the anti-mesothelin therapeutic agent, the folate antimetabolite, and/or the platinum-based chemotherapy and TIPS factor inhibitor) may be by any means known in the art.

(95) A therapeutic antibody may be used that comprises the CDR sequences that can direct binding of an antibody to the CD20 antigen: SEQ ID NO:20 (GYTFTSYN) as CDRH1, SEQ ID NO:21 (IYPGNGDT) as CDRH2, SEQ ID NO:22 (ARSTYYGGDWYFNV) as CDRH3, SEQ ID NO:23 (SSSVSY) as CDRL1, SEQ ID NO: 24 (ATS) as CDRL2 and SEQ ID NO:25 (QQWTSNPPT) as CDRL3, numbered according to IMGT. For example but not limited to, a patient may have a CD20-expressing cancer such as Hodgkin's, Non-Hodgkin's or follicular lymphoma. Previous reports have shown that anti-CD20 antibodies are bound by TIPS factors, making a TIPS factor inhibitor a desirable entity.

(96) A patient with CD20-expressing cancer that expresses a TIPS factor may be treated with a TIPS factor inhibitor. An anti-CD20 therapeutic agent may be administered to a subject along with the TIPS factor inhibitor. The TIPS factor may be CA125. An anti-CD20 therapeutic agent may be administered to the subject, where the subject has a baseline CA125 level that is above the normal

range. The method may involve administering the TIPS factor inhibitor alone. Alternatively, the TIPS factor inhibitor may be administered in combination with chemotherapy. The chemotherapy may be the combination of CHOP (cyclophosphamide, doxorubicin (hydroxydaunomycin), vincristine (Oncovin®), and prednisolone) or any other chemotherapeutic agents and/or medical procedure considered standard of care at the time when the subject is treated. CA125 expression level may be determined by any means known in the art.

(97) Various delivery systems may be used to administer the therapeutic agents (including the anti-mesothelin or anti-CD20 therapeutic agent, the folate antimetabolite, the platinum-based, and/or CHOP chemotherapy or TIPS factor inhibitor) including intradermal, intramuscular, intraperitoneal, intravenous, subcutaneous, intranasal, epidural, and oral routes as deemed necessary. The agents may be administered, for example by infusion or bolus injection, by absorption through epithelial or mucocutaneous linings (e.g., oral mucosa, rectal and intestinal mucosa, and the like) via systemic or local approaches.

(98) The therapeutic agents may be administered by injection via syringe, catheter, suppository, or any implantable matrix or device.

(99) The therapeutic agents and pharmaceutical compositions may be administered orally in any acceptable dosage form such as capsules, tablets, aqueous suspensions, solutions or the like.

(100) Preferred methods of administration of the therapeutic agents include but are not limited to intravenous injection and intraperitoneal administration at a final concentration suitable for effective therapy.

(101) The TIPS factor inhibitor in combination with other drugs may be administered as pharmaceutical compositions comprising a therapeutically or prophylactically effective amount of the therapeutic agent(s) and one or more pharmaceutically acceptable or compatible ingredients.

(102) The amount of the therapeutic agent that is effective in the treatment or prophylaxis of a cancer may be determined by standard clinical techniques. In addition, in vitro assays may optionally be employed to help identify optimal dosage ranges required for TIPS factor inhibitor. Effective doses may be extrapolated from dose-response curves of TIPS factor inhibitor derived from in vitro or animal model test systems.

(103) For example, toxicity and therapeutic efficacy of the agents can be determined in cell cultures or experimental animals by standard pharmaceutical procedures for determining the LD.sub.50 (the dose lethal to 50% of the population) and the ED.sub.50 (the dose therapeutically effective in 50% of the population) values. The dose ratio between toxic and therapeutic effects is the therapeutic index and it may be expressed as the ratio LD.sub.50/ED.sub.50. Agents that exhibit large therapeutic indices are preferred. When an agent exhibits toxic side effects, a delivery system that targets the agent to the site of affected tissue may be used to minimize potential damage to non-mesothelin-expressing cells and, thereby, reduce side effects.

(104) The dosing and dosage schedule may vary depending on the active drug concentration, which may depend on the needs of the subject.

(105) Composition of Kits to Identify Tips Factors and Tips Susceptible Antibodies (TSA)

(106) Further provided here are kits for making complementary probes to screen antibodies to determine their dynamic structure in the presence and absence of potential test antibody binding TIPS factors and for screening to identify TIPS factor inhibitors. The kits may contain a labeled fluorescent Protein L (SEQ ID NO: 1) labeled with a fluor, such as the one listed within this document, and a Protein A (SEQ ID NO: 2) or Protein G (SEQ ID NO:3) labeled with a complementary fluor that can be stimulated by the fluor bound to Protein L, a vessel for containing the Protein L, Protein A and/or Protein G when not in use, and instructions for using the paired probes for determining if a TIPS factor can affect the dynamic structure of a dual-labeled test antibody. The kit may have instructions on how to conjugate the Protein L and Protein A probes to effectively generate a dual-labeled test antibody. The instructions may specify that a baseline level of signal as determined by a fluorescent reading instrument and that a signal that is altered by at

least 8% is indicative of the TIPS having an effect on test antibody, and said antibody being a TIPS Susceptible Antibody (TSA). Instructions may specify that a baseline signal change of 50% is indicative of a TIPS factor having an effect on test antibody. Alternatively, the instructions may specify that the probes may be added to 384 or 96 well microtiter plate formats to screen multiple TIPS factors simultaneously, wherein the TIPS factor may be from a biological fluid, primary or cultured tumor cells or an isolated protein or non-protein agent. Microtiter plates may also contain libraries of peptides, proteins, nucleic acids, natural products, and/or small chemical agents (here referred to as compounds) whereby the probes are added to a TSA in the present of TIPS factors and added to 384 or 96 well microtiter plates containing a said library to identify a compound that can block TIPS factor effect on TSA dynamic structure. A signal that is reduced or enhanced 8% or greater is indicative of a compound having TIPS factor inhibitor activity. One or more additional containers may enclose elements, such as reagents or buffers, which may be used in the assay(s). Such kits can also, or alternatively, contain a detection reagent that contains a reporter group suitable for direct or indirect detection of TIPS factor antibody binding.

(107) Kits for making complementary aptamer-based probes may be used to screen test antibodies to determine their dynamic structure in the presence and absence of TIPS factor and for screening TIPS factor inhibitors. The kits contain a labeled fluorescent aptamer that can bind to an antibody Fab domain and not disrupt its ability to bind its antigen. The fluor can be any fluor, such as one from the list of fluors listed within this document, and a second aptamer that can specifically bind to an antibody Fc domain labeled with a complementary fluor that can be stimulated by the fluor bound to the Fab aptamer, a vessel for containing the Fab and Fc binding aptamers when not in use, and instructions for using the paired probes for determining if a TIPS factor can affect the dynamic structure of a dual-labeled test antibody. The instructions may include methods on how to chemically conjugate the aptamers to create an effective dual labeled test antibody. The instructions may specify that a baseline level of signal as determined by a fluorescent reading instrument and that a signal that is altered by at least 8% is indicative of TIPS factor having an effect on test antibody, and said antibody being a TSA. Instructions may specify that a baseline signal change of 50% is indicative of TIPS factor having an effect on test antibody. Alternatively, the instructions may specify that the probes may be added to 384 or 96 well microtiter plate formats to screen multiple TIPS factors simultaneously, wherein the TIPS factors may be from a biological fluid, primary or cultured tumor cells or an isolated protein or non-protein agent. The TIPS factor may be known or unknown. The microtiter plates may also contain libraries of peptides, proteins, nucleic acids, natural products, and/or small chemical agents (here referred to as compounds) whereby the probes are added to a TSA in the present of TIPS factor and added to 384 or 96 well microtiter plates containing a said library to identify a compound that can block TIPS factor effect on TSA dynamic structure where a signal that is reduced or enhanced 8% or greater is indicative as a TIPS factor inhibitor. One or more additional containers may enclose elements, such as reagents or buffers, to be used in the assay(s). Such kits can also, or alternatively, contain a detection reagent that contains a reporter group suitable for direct or indirect detection of TIPS factor antibody binding.

(108) Kits may contain a labeled fluorescent antibody that can bind to a test antibody Fab domain and not disrupt its ability to bind its antigen. The fluor may optionally be selected from a list of fluors listed within this document. The kit may also comprise a second antibody that can specifically bind to an antibody Fc domain labeled with a complementary fluor that can be stimulated by the fluor bound to the Fab binding antibody, a vessel for containing the Fab and Fc binding antibodies when not in use, and instructions for using the paired probes for determining if a TIPS factor can affect the dynamic structure of a dual-labeled test antibody. The instructions may include optimal methods for chemically conjugating the anti-Fab and -Fc antibodies to test antibody to generate an effective dual-labeled test antibody probe. The instructions may specify that a baseline level of signal as determined by a fluorescent reading instrument and that a signal

that is altered by at least 8% is indicative of the test antibody being a TSA affected antibody. Instructions may specify that a baseline signal change of 50% is indicative of a TSA antibody. Alternatively, the instructions may specify that the probes may be added to 384 or 96 well microtiter plate formats to screen multiple TIPS factors simultaneously, wherein the TIPS factor may be from a biological fluid, primary or cultured tumor cells or an isolated protein or non-protein agent. It may be known or unknown. The microtiter plates may also contain libraries of peptides, proteins, nucleic acids, natural products, and/or small chemical agents (here referred to as compounds or agents) whereby the probes are added to a dual-labeled test TSA antibody in the present of TIPS factors and added to 384 or 96 well microtiter plates containing a said library to identify a compound that can block TIPS factor effect on TSA dynamic structure where a signal that is reduced or enhanced 8% or greater is indicative as a TIPS factor inhibitor. One or more additional containers may enclose elements, such as reagents or buffers, to be used in the assay(s). Such kits can also, or alternatively, contain a detection reagent that contains a reporter group suitable for direct or indirect detection of TIPS factor antibody binding.

(109) The kits may contain a labeled fluorescent antigen that is specifically bound by the test antibody in such a way that the label does not disrupt the test antibody's ability to bind to its antigen and is labeled with a fluor such as one listed within this document and a second probe that binds to an antibody Fc domain labeled with a complementary fluor that can be stimulated by the fluor bound to the antigen, a vessel for containing the antigen and Fc probe when not in use, and instructions for using the paired probes for determining if a TIPS factor can affect the dynamic structure of a dual-labeled test antibody. The Fc probes can be from a number of known Fc binding proteins including but not limited to human FCGR1A (SEQ ID NO:4), FCGR2A (SEQ ID NO: 5), FCGR2B (SEQ ID NO:6), FCGR2C (SEQ ID NO:7), FCGR3A (SEQ ID NO: 8), FCGR3B (SEQ ID NO:9), FCGRT (SEQ ID NO:10), FCRL5 (SEQ ID NO: 11), mouse FCGR4 (SEQ ID NO:12), human C1q (SEQ ID NO:13) and/or derivatives thereof, a Fc domain specific aptamer or Fc specific antibody or antibody fragment, Protein A or G. The instructions may specify the optimal means to chemically conjugate probes to test antibody to make an effective dual-labeled test antibody. The instructions may specify that a baseline level of signal as determined by a fluorescent reading instrument and that a signal that is altered by at least 8% is indicative of TIPS factor having an effect on a TSA. Instructions may specify that a baseline signal change of 50% is indicative of TIPS factor having an effect on test antibody. Alternatively, the instructions may specify that the probes may be added to 384 or 96 well microtiter plate formats to screen multiple TIPS factors simultaneously, wherein the TIPS factors may be from a biological fluid, primary or cultured tumor cells or an isolated protein or non-protein agent. The microtiter plates may also contain libraries of peptides, proteins and/or small chemical agents (here referred to as compounds) whereby the probes are added to a dual-labeled test TSA in the present of TIPS factors and added to 384 or 96 well microtiter plates containing a said library to identify a compound that can block TIPS factor effect on TSA dynamic structure where a signal that is reduced or enhanced 8% or greater and does not affect the antigen binding to test antibody is indicative as a TIPS factor inhibitor. One or more additional containers may enclose elements, such as reagents or buffers, to be used in the assay(s). Such kits can also, or alternatively, contain a detection reagent that contains a reporter group suitable for direct or indirect detection of TIPS factor antibody binding.

(110) Kits may also comprise a mixture of Fab and Fc probes, for example using fluor labeled antigen in combination with fluor labeled Protein A, Protein G, an Fc specific antibody or antibody fragment, or a Fc binding protein. Another example is using a fluor labeled Protein L in combination with a fluor labeled Fc specific antibody or antibody fragment, or a Fc binding protein in conjugate or non-conjugate form. Yet another example may be a fluor labeled aptamer in combination with a complementary fluor labeled Fab or Fc binding protein.

(111) Kits may also comprise an unlabeled complementary Fab probe (Protein L, aptamer, antigen, etc.) that can be used in ELISA format to capture test antibody and a second biotin or other detector

capable label and unlabeled probe that can bind to test antibody Fc domain (Protein A, Protein G, Fc receptors, etc). This format is used for primary screening of TIPS factors to determine if they have an effect on Fc binding probes by: (1) coating ELISA plate with Fab probe, (2) adding test antibody, with or without TIPS and Fc-biotin probe to measure binding in presence or absence of TIPS factor. If no difference in Fc probe binding is observed between TIPS factor treated and control wells, then said probes are suitable for pairing. Finally, kit may contain biotinylating agent to label test antibody to determine ability of test antibody to bind antigen in the present of unlabeled Fab and Fc probes by: (1) coating wells with test antibody's antigen, (2) adding Fab and Fc probe to biotinylated test antibody and screening for antigen binding via secondary streptavidin-HRP. If no difference exists between antigen binding in presence of Fab and Fc probes as compared to test antibody alone, the probes are sufficient as pairs to test against TIPS factors and determine if test antibody is a TSA.

(112) Kits also typically contain a label or instructions for use in the methods described here. The label or instruction refers to any written or recorded material that is attached to, or otherwise accompanies a kit at any time during its manufacture, transport, sale or use. It can be a notice in the form prescribed by a governmental agency regulating the manufacture, use or sale of pharmaceuticals or biological products, which notice reflects approval by the agency of manufacture, use or sale for human administration. The label or instruction can also encompass advertising leaflets and brochures, packaging materials, instructions, audio or videocassettes, computer discs, as well as writing imprinted directly on the pharmaceutical kits.

(113) The above disclosure generally describes the present invention. All references disclosed here are expressly incorporated by reference. A more complete understanding can be obtained by reference to the following specific examples, which are provided here for purposes of illustration only, and are not intended to limit the scope of the invention.

Example 1—Developing Complementary Binding Pairs to Monitor for Tips Factor Binding to Dual-Labeled Test Antibody and Effect on their Dynamic Structure

(114) FIG. 1 shows a schematic diagram of an antibody structure and approximate regions where Fab/N-terminal and Fc/C-terminal specific probes may bind and be tested for their ability to be used in pairs to measure antibody dynamic structure. While many of these proteins have been shown to bind antibodies, the ability to use two simultaneously to monitor antibody dynamic structure in a high throughput and functional based assay is unknown. In particular, we teach methods on the how to identify and combine probes that may remain bound to a dual-labeled test antibody and not be displaced by a TIPS factor using chemical conjugation and screening to ensure such treatment does not affect test antibody binding to its antigen.

(115) FIG. 2 provides an example of the ability of probe pairs to be used in combination with a dual-labeled test antibody by simultaneous binding without affecting antibody binding to target antigen. As shown, while some probes are compatible for simultaneous use (i.e., Protein L and Protein G or Protein A; antigen and FCGR1A), others are not (i.e., Protein L and C1q, test antibody's antigen and FCGR2A or FCGR3A), highlighting the importance of kits with defined pairs of compatible probes to measure antibody dynamic structure.

(116) To test for compatible Fab and Fc binding pairs to said test antibody in the presence or absence of TIPS factors, probes were applied to an ELISA format using a Fab probe as capture and a Fc probe as detector. In cases where the two probes could bind test antibody in the presence of TIPS factor, an additional test was run to ensure that the probe-bound test antibody was still capable of binding its target antigen. Tests of Fab and Fc probes consisting of antibodies, antibody fragments, Fc binding proteins FCR1A, FCR2A, FCR3A, FcRn, C1q, Protein L, Protein A and Protein G were carried out in ELISA format using buffers as previously described (Kline J B, et al. Oncotarget 8:52045-52060, 2017) with the exception of probing and washing for FcRn, which was done using powdered milk and PBS buffer pH 5.5. Briefly, 96-well plates were coated with 1 µg/mL of each capture probe, test antibody or antibody specific antigen overnight in 0.05M

carbonate buffer at 4° C. The next day wells were rinsed with PBS (pH 7.2) and blocked with 5% BSA in PBS for 1 hour, then washed three times in PBS. Next, test antibodies (rituximab and pertuzumab were used as positive and negative controls, respectively), [TIPS factor CA125 has been reported to bind rituximab but not pertuzumab (Kline J B et al. *Eur J Immunol* 48:1872-1882, 2018)] with or without CA125 and a biotinylated Fc probe were added and plates were incubated for 1 hr at room temp. Next, wells were washed in PBS and secondarily probed with streptavidin-horseradish peroxidase (SRTP-HRP) for 1 hr at room temp. Wells were then washed 3 times and TMB substrate (ThermoScientific) was added for 15 min. Reactions were stopped with 0.1N H.sub.2SO.sub.4 and plates were read at 450 nm optical density (OD) using a 96-well Varioscan plate reader. As shown in FIG. 2, the use of Protein L capture and FCGR2A, FCGR3A and Protein G probes were compatible as simultaneous Fab and Fc antibody binding was consistent and robust and test antibody was able to also bind antigen when in combination with both probes. This was demonstrated by coating ELISA plate with target antigen (CD20 peptide for rituximab and HER2 antigen for pertuzumab) and adding biotinylated rituximab or pertuzumab in the presence of Fab and Fc unlabeled proteins. Protein L and C1q pairs were not compatible. The ability of CA125 to perturb pair probe binding using Protein L as capture and biotinylated FCGR1A, FCGR2A, FCGR3A or Protein A or G as detector in the presence or absence of CA125 was then tested. As shown in FIG. 3, CA125 resulted in decreased FCGR3A binding but not FCGR1A, Protein A (not shown) or Protein G to rituximab. No effect was seen for pertuzumab as expected as CA125 does not bind pertuzumab (Kline J B, et al. *Oncotarget* 8:52045-52060, 2017; Gunn, et al. *Mucosal Immunol* 9:1549-1558, 2016). These data demonstrate that finding complementary probes that can simultaneously bind a test antibody for detection is required but not sufficient to be useful for identification of TIPS factor Sensitive Antibodies (TSAs) such as rituximab. Probes may have varying, unobvious behaviors when a test antibody interacts with a TIPS factor, making these probes less effective as screening tools to identify TSAs as well as TIPS factors and for screening for TIPS factor inhibitors. Pairs that are more stable (i.e. Protein L and Protein G; Protein L and Protein A; test antibody antigen and FCGR1A, or other probes including complementary antibody and/or aptamers Fab and Fc binding probes) may be most useful in studying the antibody dynamic structure as described and taught in Example 2 in the presence of TIPS factors.

(117) As shown in FIG. 3, TIPS factors such as CA125 that can bind to a TSA such as rituximab, can lead to altered dynamic structure and antibody humoral function. These assays are useful for monitoring functional effects that TIPS factor and TIPS factor inhibitors may have on a dual-labeled test antibody. All biotinylated probes were created using sulfo-tag conjugation as described by the manufacturer (Meso Scale Diagnostics Inc). All reactions were done in at least triplicate and mean values were analyzed using students T-test.

Example 2—Determination of Complementary Probes to Measure Antibody Dynamic Structure of Dual-Labeled Test Antibody in the Presence or Absence of TIPS Factors

(118) As shown in Example 1, it is possible to have probes bind the N-terminus/Fab domain and the C-terminus/Fc domain of a test antibody and then measure the effects of a TIPS factor to bind and alter the test antibody's dynamic structure resulting in decreased binding of low affinity Fc receptors such as FCGR2A and FCGR3A but not higher affinity binding proteins (i.e., Protein L, Protein G, FCGR1A), antibodies and/or aptamers. The latter group of probes can retain binding to test antibody in the presence of TIPS factor binding to test antibody and are useful in additional formats that can monitor the dynamic structure of test antibody by measuring its N- and C-terminal distance at steady-state and in presence of a TIPS factor to determine if the TIPS factor binds and alters the test antibody's dynamic structure. If the distance is altered by a TIPS factor, then the antibody is determined to be bound by that TIPS factor and its dynamic structure is likely altered as is the case of rituximab and CA125 regarding the decreased FCGR2A (not shown) and FCGR3A binding.

(119) As taught here, the ability to test antibody dynamic structure to TIPS factor binding in a high

throughput system offers a unique opportunity to identify those antibodies that are susceptible to TIPS factors (TIPS Susceptible Antibody, TSAs) and allows during development of a therapeutic antibody program: (1) the avoidance of patients that express a said TIPS factor to be treated with an affected TSA; (2) use of the paired probe system to screen for compounds that can overcome the susceptibility of TSA to TIPS factor(s) binding; and/or (3) the use of dual-labeled test antibodies with complementary fluorescent probes that can monitor the steady-state distance between the N- and C-terminal domains in the presence or absence of TIPS factor to either engineer a refractory TSA or develop a TIPS factor inhibitor. Any method of labeling a dual-labeled test antibody as described above can be used along with complementary fluors that can be monitored via fluorescent resonance energy transfer (FRET) in which an increased FRET signal would indicate the TIPS factor causing the test antibody's N and C-terminal ends to become closer in proximity while a decreased FRET signal would indicate the TIPS factor causing them to become more distal; in either case indicating an effect on test antibody's dynamic structure and function.

(120) As shown in FIG. 4, a dual-labeled test antibody was formed by the binding of fluor labeled antibody probes that could specifically recognize the Fab and Fc domains of a test antibody. The anti-Fab antibody was labeled with Alexa fluor AF488 and the anti-Fc antibody was labeled with Alexa fluor AF555. Probes were simultaneously crosslinked to the test antibody using 2.5 mM disuccinimidyl suberate in 100 mM NaHPO_{sub.4}, 20 mM HEPES, 150 mM NaCl and 100 mM H_{sub.2}CO_{sub.3}/Na_{sub.2}CO_{sub.3} followed by quenching with 1M Tris-HCl, pH 7.5. Samples were desalted and purified over a protein A column, which binds the humanized test antibody but not the non-human conjugated antibody probes. The test antibody was analyzed for concentration and dual labeling via NanoDrop One™ spectrophotometry/fluorometry as recommended by the manufacturer (Thermo Scientific). As shown in FIG. 4A, a dual-labeled AF488 and AF555 test antibody was generated and used in a fluorescent resonance energy transfer (FRET) assay shown in FIG. 4B. Briefly, the Fab488-Fc555 dual-labeled or individual-labeled test antibodies were incubated with 10 ng/ml of TIPS factor CA125, human serum albumin (HSA) or buffer only in 0.05M phosphate buffer, pH 7.0 in a black 96-well opaque microplate (Greiner) and quantified via FRET using a Varioscan fluorescent plate reader in triplicate. Samples were incubated from 10 min to 3 hrs and tested for FRET activity using 470 nm excitation with a delay time of 0.05 msec, an integration time of 0.4 msec and read at an emission wavelength of 570 nm following methods previously described (Chakraborty S, et al. PLOS ONE 9:e102572, 2014). As shown, HSA had a similar value as probe in buffer alone, while the dual-labeled test antibody treated with CA125 had a 35% increase in 570 nm emission signal demonstrating an increase in FRET signal due to the N- and C-termini coming into closer proximity. These values were similar among timepoints of the experiment and represent a minimum of triplicate datapoints. The addition of single-labeled test antibody with Fab555 plus single-labeled test antibody with Fc488 resulted in no enhanced FRET signal when CA125 was added and had an overall much lower baseline signal in buffer alone than the dual-labeled antibody, 1.850 vs 4.880 at 570 nm, demonstrating the need for the two fluor dyes to be in close proximity and on the same molecule for detecting TIPS factor effect(s) on an antibody's dynamic structure.

(121) FIG. 5, demonstrates the use of a dual-labeled test antibody formed by the binding of AF555-labeled Protein L (referred to as PL555) that specifically recognizes the Fab light chain and AF488-labeled Protein A (referred to as PA488) that recognizes the test antibody's Fc domain. The generation of PA488 proteins have been previously reported (ThermoFisher.com/secondary). No previous findings have identified the attempted development of an AF555-labeled Protein L for any utility including its use as a secondary probe for AF488 paired FRET. Here we teach the generation and utility of PL555 for FRET applications. To generate PL555, 1.9 mM of recombinant Protein L (Pierce Biotechnology) was incubated with 750 μM succinimidyl ester labeled AF555 in 0.1M Na_{sub.2}CO_{sub.3} for 1 hr. Reactions were desalted using 7 kDa molecular weight cut off (7-MWCO) Zeba columns (ThermoScientific) and Protein L-AF555 labeling was confirmed by SDS-

PAGE electrophoresis/UV visualization via imaging on an I-Bright CL1000 imager (ThermoFisher) and NanoDrop One spectrophotometry/fluorometry. As shown in FIG. 5A, an AF555-Protein L labeled probe with the correct fluor spectrum of 555 nm (left figure) and molecular weight of 36 kDa (right figure) was generated. Next, a 2:1 molar volume of test antibody and PL555 and PA488 together (for dual labeling) or each alone (for single labeling) were incubated in the presence of 2.5 mM DSS as described above. Test antibodies were purified via size exclusion using a 100 kDa molecular weight cutoff (100-MWCO) dialysis tube in PBS (pH 7.0) for 72 hrs at 4° C. Antibodies were then analyzed for concentration and dual labeling as described above. Testing for antigen binding demonstrated that the dual-labeled test antibody still retained antigen binding (not shown). FIG. 5B, demonstrates that the dual-labeled test antibody was generated and was next used in fluorescent resonance energy transfer (FRET) assay using methods as described above. Results are shown in FIG. 5C. Briefly, the PL555-PA488 dual-labeled test antibody was incubated with 10 ng/ml of TIPS factor CA125, human serum albumin (HSA) or buffer only in 0.05M phosphate buffer, pH 7.0 and quantified via FRET using a 96-well Varioscan fluorescent plate reader. Samples were incubated 10 min to 3 hours and tested for FRET activity using a 470 nm excitation wavelength with a delay time of 0.05 msec, an integration time of 0.4 msec and read at an emission wavelength of 570 nm. As shown, HSA had similar value as probe in buffer alone, while test probe treated with CA125 had a 18% increase in 570 nm emission signal, demonstrating an increase in FRET signal in the presence of CA125. These values were similar among timepoints of the experiment and represent a minimum of triplicate datapoints. The addition of single-labeled test antibody labeled with PL555 plus single-labeled test antibody labeled with PA488 resulted in no FRET signal when CA125 was added and had a much overall lower baseline 0.655 vs 4.70 at 570 nm, demonstrating the need for the two complementary fluors to be in close proximity on the same molecule to detect TIPS factor binding and effect on a test antibody's dynamic structure.

(122) As shown in FIG. 6, a dual-labeled test antibody was formed by direct conjugation to unique residues within a test antibody. The anti-CD20 antibody rituximab has been shown to be bound by CA125 that in turn suppresses its humoral immune function(s) (Kline J B et al. Eur J Immunol 48:1872-1882, 2018). Recent mapping localized the CA125 binding to the Fab region. We engineered a variant rituximab containing an unpaired cysteine (CYS) at residue 169 in the heavy chain (S169C) using a standard PCR-based random mutagenesis of the Fab domain regions. Randomized fragments were cloned into eukaryotic expression vectors, transiently transfected into mammalian host cells and functionally screened using culture supernatants to identify putative rituximab mutants that could still bind CA125 and its CD20 antigen. After screening and sequencing of hundreds of clones that retained CD20 and CA125 binding, we identified one clone that comprised the S169C mutant (SEQ ID NO: 26 and 27). This clone is referred to as RTX-169. After scale up of antibody production via larger scale transient transfection of 293 cells and protein A column purification, we prepared RTX-169 for direct cysteine labeling following methods as previously described (Banks D D, et al. J Pharm Sci 97:764-779, 2008). First, 1 mg/mL of rituximab wild type (RTX-WT) and RTX-169 were incubated with 1.5 mM L-cysteine (SigmaAldrich) in phosphate buffer saline, pH 7.0 overnight at 4° C. The next day, reactions were desalted using 7-MWCO Zeba columns and quantified for protein content via NanoDrop One. Each antibody was then tested for decysteinylation via the Ellman assay following the manufacturer's instruction (Gold Biotechnology). Analysis found a 12% enhancement of Ellman reagent labeling of treated vs untreated RTX-169 mAbs, while no difference was observed between treated or untreated RTX-WT, suggesting the presence of the free cysteine 169 within RTX-169. Both RTX-WT and RTX-169 were then treated with a 20:1 molar ratio of NaN.sub.3:antibody overnight at room temp as previously described (Li, X Y. Biotechnol Prog 28:856-861, 2012) and equal molar concentrations of NaN.sub.3-treated and untreated antibodies were incubated with 200 µM of dibenzylcyclooctyne-PEG4-biotin (DBCO-biotin) (Sigma Aldrich) for 1 hr at room temp to form

antibody-biotin conjugates via click chemistry, a process known by those skilled in the art. Samples were desalted using 7-MWCO Zeba columns as above and tested for biotin labeling via ELISA and western blot. For ELISA, triplicate wells were plated with 0.2 mg/mL of NaN.sub.3-treated or untreated, DBCO-biotin exposed RTX-WT or RTX-169 in 0.05M carbonate buffer overnight at 4° C. Wells were washed with 0.05M phosphate buffer, pH7.2 (PB buffer), blocked with 5% bovine serum albumin (BSA) in PB buffer then probed with a 1:3000 dilution of streptavidin-horseradish peroxidase (SRTP-HRP) for 1 hr at room temp. Wells were then washed with PB buffer and exposed to TMB substrate (ThermoScientific) for 5 min, stopped with 0.1N H.sub.2SO.sub.4 and read via OD at 450 nm using a Varioscan 96-well plate reader. As shown in FIG. 6A, The RTX-169 had a significantly enhanced biotin labeling than the either the NaN.sub.3-untreated RTX-169 or the NaN.sub.3-RTX-WT treated antibodies ($P < 0.000009$), which was likely due to the free cysteine at residue 169 within RTX-169. To confirm labeling in the N-terminal Fab domain, RTX-169 was digested with papain and Fab and Fc fragments isolated via Protein L and Protein A agarose beads as recommended by the manufacturer (ThermoScientific). Full length RTX-169-DBCO-biotin labeled and unlabeled RTX-169 along with DBCO-biotin labeled RTX-169 Fab and Fc fragments were electrophoresed on a 4-12% non-denaturing SDS-PAGE gel, electroblotted and probed by western blot for the presence of biotin using STRP-HRP. As shown in FIG. 6B, the RTX-169-DBCO-biotin full length and Fab fragment both bound SRTP-HRP while the untreated RTX-169 and Fc-fragment of the DBCO-biotin labeled RTX-169 did not, confirming biotin specific labeling within the RTX-169-DBCO-biotin Fab domain. We next used the DBCO-biotin tag on RTX-169 to label RTX-169 with Alexa fluor AF555. RTX-169-DBCO-biotin was incubated with an avidin-labeled AF555 and PA488 at a 2:1 ratio for 1 hour in the presence of DSS as described above. Antibody was purified via dialysis using a 100-MWCO dialysis membrane and analyzed by a denaturing SDS-PAGE gel that showed high molecular weight species as expected for RTX-169-AF555-PA488 dual-labeled antibody (not shown). The test antibody was further analyzed by NanoDrop One spectrophotometry/fluorometry and confirmed to be dual labeled with the AF488 and AF555 fluors (FIG. 6C). Antigen binding of the dual-labeled antibody was confirmed via ELISA (not shown). The test antibody was then used in FRET using similar parameters as above in the presence of buffer, HSA or CA125 and quantified for 10 min to 3 hrs. As shown in FIG. 6D, the RTX-169-AF555-PA488 antibody showed a positive FRET signal when incubated with CA125 in contrast to HSA or buffer alone, confirming the utility of this format to detect TIPS factor binding to TSAs such as rituximab.

(123) The teachings of the compositions of reagents and the methods for their use to uncover TSAs and TIP factors enables the utility to use specific binding of probes such as but not limited to anti-Fab or anti-Fc antibodies, Protein A, G or L binding proteins, Fab and/or Fc binding aptamers as well as single amino acid changes within the Fab and/or Fc domains to form dual-labeled test antibodies. The formation of such dual-labeled test antibodies using any combination of complementary fluor pairs or other detector agents as discussed above is suitable for identifying potential TIPS factors that can bind test antibodies, identifying TSA affected antibodies, and their use for screening chemical and protein libraries for TIPS factor inhibitors as novel therapies as well as using TIPS expression profiles for patient selection.

Claims

1. A method for characterizing an antibody comprising a Fab and a Fc domain, comprising: contacting a first fluor-labeled protein L (SEQ ID NO: 1), and a second fluor-labeled protein A (SEQ ID NO: 2), with the antibody to be characterized to form a ternary complex, wherein the first fluor-labeled protein L binds to the Fab domain and the second fluor-labeled protein A binds to the Fc domain, wherein the first and second fluors participate in FRET, wherein the antibody to be characterized is an anti-tumor antibody obtained from a cancer patient; determining fluorescence

resonance energy transfer (FRET) of the ternary complex; contacting the ternary complex or components of the ternary complex with tumor-induced or -produced factor (TIPS factor) Cancer Antigen 125 (CA125); and determining FRET of the ternary complex in the presence of the TIPS factor.

2. A method of screening test substances for the ability to mitigate an effect of tumor-induced or -produced factor (TIPS factor) Cancer Antigen 125 (CA125) on TIPS-susceptible antibody selected from the group consisting of rituximab, trastuzumab, cetuximab, YP219, trastuzumab emtansine, ocrelizumab, daratumumab, elotuzumab, alemtuzumab, necitumumab, pertuzumab, obinutuzumab, nivolumab, ipilimumab, pembrolizumab, ofatumumab, panitumumab, ibritumomab tiuxetan, sacituzumab govitecan, brentuximab vedotin and tositumomab, comprising a Fab domain and a Fc domain, the method comprising: contacting the TIPS-susceptible antibody with (a) a first fluor-labeled protein L (SEQ ID NO: 1), that specifically binds to the Fab domain of the TIPS-susceptible antibody, and (b) a second fluor-labeled protein A (SEQ ID NO: 2) that specifically binds to the Fc domain of the TIPS-susceptible antibody, to form a first complex; contacting the first complex with (c) the TIPS factor, to form a second complex; contacting at least some of the second complex with (d) a test substance; and measuring fluorescence resonance energy transfer (FRET) of the first complex, the second complex in the absence of the test substance, and the second complex in the presence of the test substance.
